
SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Spatial reasoning is a fundamental pillar of human cognition, yet it remains an unsolved challenge for contemporary vision–language models (VLMs). Prior work relies on synthetic or LLM-generated environments with narrow task designs and puzzle-like setups, and consequently fails to capture the real-world complexity, visual noise, and diverse spatial relationships that VLMs must handle in practice. We close this gap with **SPATIALAB**, a comprehensive benchmark that evaluates VLMs’ spatial reasoning in realistic, unconstrained contexts. SPATIALAB comprises 1,400 visual question–answer pairs across six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each with five subcategories, yielding 30 distinct task types. With at least 25 questions per subcategory and at least 200 per main category, the benchmark delivers balanced coverage and drives both multiple-choice and open-ended evaluation. Experiments spanning diverse state-of-the-art VLMs, including open- and closed-source models, reasoning-focused systems, and specialized spatial reasoning models, expose a wide and consistent gap between machine and human spatial competence. In the multiple-choice setup, the best model, InternVL3.5-72B, reaches only 54.93% accuracy against 87.57% for humans. In the open-ended setting, every model drops a further 10–25%, and even the strongest system, GPT-5-mini, attains just 40.93% against 64.93% for humans. These results pinpoint decisive failures in handling complex spatial relationships, depth perception, navigation, and 3D geometry. By delivering a diverse, real-world evaluation framework, SPATIALAB lays bare the critical challenges and opportunities for advancing VLMs’ spatial reasoning and establishes the benchmark that will guide future research toward robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a fundamental pillar of human cognition, encompassing the understanding of spatial layouts, the imagination and manipulation of objects, and navigation through environments. It underpins applications ranging from robotics, autonomous driving, and augmented/virtual reality to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). Artificial agents that perceive and act in dynamic environments demand reliable spatial competence: they must interpret geometric relations, depth cues, occlusion patterns, and agent-centric perspectives in scenes that are visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Recent vision–language models have markedly advanced multimodal representation and language grounding, yet their spatial judgments remain decidedly fragile in realistic environments (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks, however, are narrow and simplified. Most target elementary tasks such as binary spatial relations, low-resolution depth categories, or schematic navigation cues, almost always in synthetic or simplified settings that strip away visual clutter and homogenize objects (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). These controlled setups relax perceptual and reasoning demands, manufacturing apparent saturation while masking outright failures under distribution shift. Critical challenges, robust occlusion inference, scale-consistent judgments

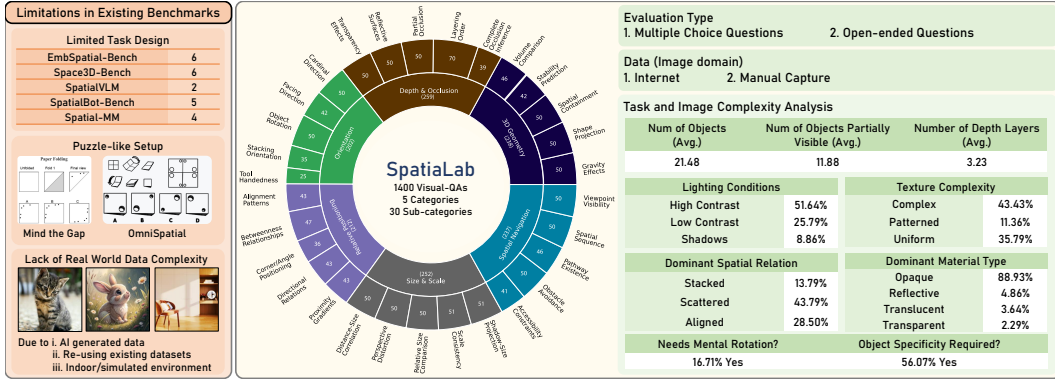


Figure 1: **Overview of SPATIALLAB.** The benchmark overcomes the limitations of prior datasets (left), contributes 1,400 visual QA pairs spanning 5 categories and 30 subcategories (center), and enables systematic evaluation through multiple-choice and open-ended tasks. It delivers pronounced task and image complexity, with varied object counts, layers, lighting, textures, relations, and materials (right).

across viewpoints, and planning under partial observability, remain badly under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). The consequence is stark: models that top existing benchmarks routinely fail on integrated, multi-step inference, blocking safe deployment in real-world environments where spatial errors carry immediate practical costs.

A meaningful evaluation must therefore span every core axis of spatial reasoning: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Human cognition integrates these dimensions seamlessly under noise and ambiguity, precisely what current task-isolated setups fail to demand (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). A robust benchmark must therefore test both choice and generation, measure directly against human performance, and draw from real-world scenes that surface the failures simulator datasets conceal. These principles drive the design of our benchmark.

To meet these needs, we introduce **SPATIALLAB**, a benchmark that assesses the spatial reasoning of vision–language models in realistic, unconstrained visual contexts, as outlined in Figure 1. **SPATIALLAB** provides a balanced suite of visual question–answer items spanning a broad taxonomy of spatial tasks and supports both multiple-choice and open-ended evaluation formats. The benchmark deliberately foregrounds photographic diversity, complex spatial relational queries, and tasks that require integrating depth, perspective, and 3D structure. We evaluate a wide collection of state-of-the-art VLMs, including open- and closed-source models and reasoning-focused systems, benchmark them against human performance, and conduct extensive error analysis. Our key contributions are as follows:

1. We introduce **SPATIALLAB**, a large-scale benchmark of 1,400 visual question–answer pairs spanning six principal categories and thirty task types in realistic, unconstrained contexts. With at least 200 questions per category and 25 per subcategory, it guarantees balanced coverage and supports both multiple-choice (SPATIALLAB-MCQ) and open-ended (SPATIALLAB-OPEN) evaluation.
2. We conduct a **comprehensive evaluation** of 25+ state-of-the-art vision–language models: open- and closed-source, reasoning-oriented, and spatially specialized reasoning models against human baselines, and demonstrate a consistent, substantial gap in spatial reasoning.
3. We deliver **in-depth error analyses and diagnostic studies** that pinpoint systematic failure modes in depth perception, navigation, occlusion, and 3D geometry, yielding concrete insights and actionable directions for advancing robust, human-aligned spatial understanding.
4. We investigate multiple strategies to enhance spatial reasoning in VLMs, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and **multi-agent architectures (SPATIOXOLVER) with multi-step reasoning**, and establish exactly where each helps and where it falls short across multiple-choice and open-ended tasks.

Table 1: SPATIALAB set against prior spatial reasoning benchmarks.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval		A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth				
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M.E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93

In the multiple-choice setting, leading models reach only 50–55% accuracy, while human annotators exceed 85%. In open-ended generation, model performance collapses by a further 10–25 percentage points, leaving the best systems near 40% against roughly 65% for humans. Error analysis traces these failures directly to occlusion inference, scale-consistency, and multi-step navigational planning, exposing clear limits in compositional spatial representations and reasoning. Our study establishes which interventions deliver robust gains, which merely amplify existing biases, and the decisive role of perceptual grounding for generalizable spatial reasoning. By supplying tasks, baseline results, and diagnostic analyses, SPATIALAB sets the foundation for progress and reproducible comparison in spatial reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

Visual-spatial reasoning, the ability to perceive, represent, and manipulate spatial relations among objects, is central to human cognition and indispensable for embodied agents that navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, we describe a scene as a set of objects $O = \{o_1, \dots, o_n\}$ with attributes (e.g., position, orientation, size) in a world coordinate system W . Visual-spatial reasoning then seeks to learn a mapping $f : (I, Q) \mapsto R$, where I is an image (or image sequence), Q is a linguistic query, and R is a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ capturing predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), learning f is a formidable challenge: it demands integrating noisy visual perception with compositional linguistic abstraction under embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Early approaches to spatial reasoning in VLMs rely on basic spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), typically situated in synthetic datasets like CLEVR (Johnson et al., 2016), structured question-answering benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025), as Table 1 makes clear¹. While convenient for controlled evaluation, these settings oversimplify spatial structure, remove environmental noise, and ignore dynamic factors such as temporal changes or viewpoint shifts. We therefore benchmark spatial reasoning in rich, real-world contexts that capture scene complexity, temporal variation, and multiple perspectives, confronting VLMs with exactly the challenges they face in embodied or interactive applications such as robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

In SPATIALAB, we organize spatial reasoning into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each linked to a distinct cognitive faculty studied in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is further decomposed into five subcategories, producing a taxonomy of thirty task types that isolate fine-grained reasoning skills, such as directional alignment, occlusion inference, or stability prediction. This classification scheme is designed to capture the full spectrum of visual-spatial reasoning challenges. By decomposing complex reasoning into measurable, testable units, we evaluate not only whether models “solve” spatial tasks but precisely

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

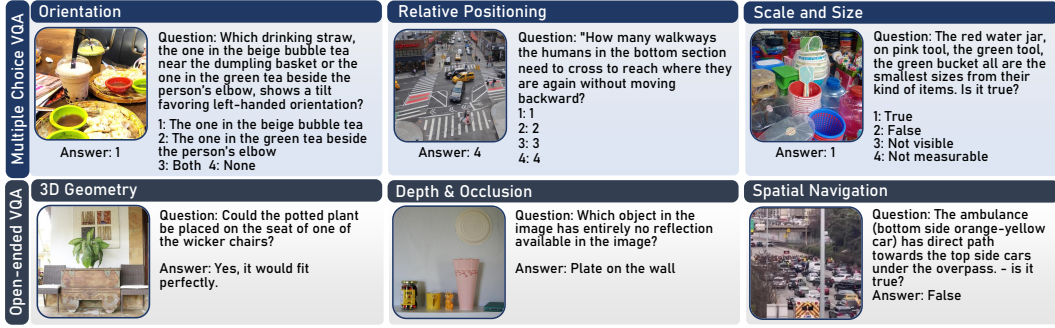


Figure 2: Representative examples spanning all six categories in the open-ended and MCQ tasks.

which dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

We further adopt two complementary evaluation formats: multiple-choice and open-ended question answering. The multiple-choice format probes recognition and fine discrimination among distractors, while the open-ended format measures generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Together, these formats mirror real-world reasoning demands and lay bare distinct strengths and weaknesses in model capabilities. By uniting a principled taxonomy, diverse real-world imagery, and dual evaluation formats, SPATIALAB delivers a structured framework for systematic benchmarking. This design enables fine-grained model comparison, direct alignment with human-level performance, and deeper insight into the development of robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains a persistent challenge for vision-language models (VLMs), most acutely in naturalistic conditions where occlusion, perspective distortion, and noisy backgrounds co-occur. We meet this challenge with SPATIALAB, a benchmark built to systematically evaluate VLMs' capacity for complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 visual question-answer pairs organized into six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each further divided into five subcategories, yielding 30 distinct task types. Every subcategory includes at least 25 QA pairs and each top-level category contributes a minimum of 200, guaranteeing balanced coverage across the taxonomy. SPATIALAB supports both multiple-choice and open-ended evaluation, assessing discriminative accuracy alongside generative reasoning and explanatory competence, and thereby aligns with broader multimodal evaluation practice while sharpening its focus on spatial-specific reasoning. High visual diversity, fine-grained annotation, and rigorous quality control establish SPATIALAB as a reliable and scalable testbed; Figure 2 presents representative examples across all six categories in both open-ended and multiple-choice formats, capturing the richness and difficulty of real-world spatial reasoning.

3.1 BENCHMARK CONSTRUCTION

Image Collection. We curated the image corpus through three complementary strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manual snapshots collected in diverse indoor and outdoor environments (Figure 3). This multi-source pipeline secures diversity across object categories, environmental conditions, and spatial configurations. To capture natural variability explicitly, we profiled every image along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Each dimension is systematically represented in the final corpus, so the benchmark spans a broad spectrum of real-world, in-the-wild scenarios rather than simplified or synthetic setups. Complexity analysis confirms the point: images contain on average 21.48 objects, of which 11.88 are partially visible, distributed across 3.23 distinct depth layers. Spatial

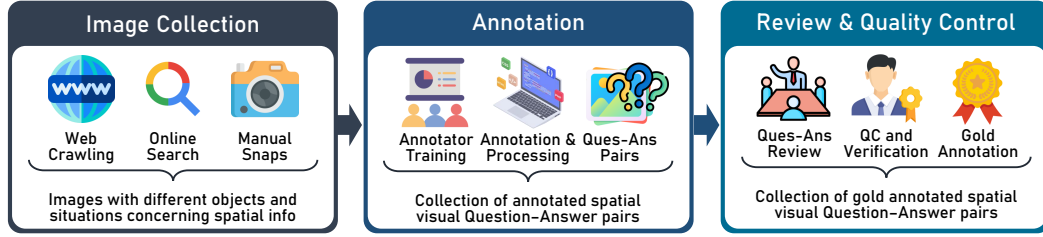


Figure 3: **Data creation pipeline for SPATIALLAB.** We collect images via web crawling, targeted search, and manual snapshots, then apply structured annotation of spatial question-answer pairs.

reference chains (e.g., “object A is left of B, which is behind C”) average 2.07 links, demanding genuine multi-step relational reasoning. Moreover, 16.71% of the questions require mental rotation and 56.07% require object specificity (a specific color, pattern, or other detail). These statistics establish that **SPATIALLAB** captures the cluttered, layered, and relationally rich scenes characteristic of real-world spatial reasoning, precisely the scenes underrepresented in synthetic datasets.

Annotation. Annotation proceeded in three phases. First, annotators underwent targeted training (details are available in Appendix C.2) to enforce consistency in interpreting spatial categories and subcategories. Second, each image was paired with one or more spatial QA tasks spanning a range of reasoning types such as proximity gradients, alignment patterns, occlusion inference, and stability prediction; questions were deliberately designed to balance perceptual grounding with higher-order inference (e.g., predicting stability under gravity constraints). Third, we encoded every QA pair in both multiple-choice (04 options) and open-ended formats, guaranteeing comparability across evaluation paradigms. This pipeline produced around 1,500 spatial visual QA pairs across six categories and 30 subcategories.

Review and Quality Control. To secure reliability, annotation outputs passed through a three-tier review and validation pipeline. In the first stage, we checked every QA item for semantic validity and alignment with its designated subcategory. In the second stage, independent annotators verified the correctness of both the question phrasing and the designated answer, eliminating ambiguity and enforcing task clarity. In the third stage, a gold-standard annotation round established a set of high-confidence QA pairs for evaluation. We removed all metadata from the images and carefully curated the dataset to avoid copyright concerns, though minor oversights may still occur. This procedure yielded a final benchmark of 1,400 rigorously validated QA items.

4 EXPERIMENTS AND EVALUATION

Models. We categorize the evaluated models into four groups. **Proprietary models** include GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** comprise Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we report open-ended evaluation only, because these models are not instruction-tuned and were trained primarily on open-ended tasks, which drives excessive refusals during MCQ evaluation. For ease of comparison in our results tables, we use color coding: proprietary, open-source up to 7B, larger than 7B, proprietary reasoning, open-source reasoning, spatial specialists, and human baseline.

Evaluation Process and Metrics. We adopt accuracy as the primary metric for both MCQ and open-ended tasks. For **MCQ evaluation**, we use direct prompting: models return the option number of their selected answer, and we automatically check every response against the correct answer. Full prompts and details appear in Appendix D.1. For **open-ended evaluation**, we prompt models to

Table 2: Multiple Choice Evaluation Accuracy (%) (↑) on SPATIALAB-MCQ by Question Categories.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

generate free-form answers and assess them with a large language model judge. Prompts and details are available in Appendix D.2. To validate the LLM judge against humans, we also carried out an LLM-human agreement analysis, detailed in Appendix D.2. Table 5 establishes that the LLM judge (Gemini-2.5-Flash) reaches substantial agreement with human annotators, with a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority vote is 0.880, and Fleiss’ kappa among human annotators is 0.774, confirming strong reliability. For additional details on experiment setup, prompts, and evaluation procedures, see Appendix D.

Improving Visual Reasoning Capabilities. To push spatial reasoning on our demanding benchmark, we investigate a full spectrum of strategies: inherent reasoning in the VLMs (Appendix H.1), Chain-of-Thought (CoT) (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT specifically, we fine-tuned Qwen-VL2.5-3B-Instruct on 40% of the dataset with stratified sampling, improving generalization across relational and geometric tasks. Evaluation on the remaining 60% shows reduced errors in navigation and orientation, with additive gains when CoT and SFT are combined. Agentic reasoning, however, benefits orientation substantially while leaving other categories to stagnate or degrade, making it clear that deeper perceptual and spatial grounding, not more multi-step reasoning, is what these tasks demand.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports model performance in multiple-choice evaluation on SPATIALAB-MCQ. Performance varies sharply across models, tasks, and families: overall accuracies span only 30–55% (random choice = 25%) against a human baseline of 87.57%, leaving vast headroom. The best model, the open-source InternVL3.5-72B, reaches just 54.93%. Reasoning- and instruction-tuned models (e.g., o4-mini: 53.21%, Gemini-2.5-Flash-Thinking: 52.93%, GPT-5-mini: 54.29%) and a few very large open-source models (e.g., InternVL3.5-72B: 54.93%, Llama-3.2-90B: 50.36%) rank near the top, but scale alone does not decide the outcome: Llama-3.2-11B, for instance,

Table 3: Open-ended Evaluation Accuracy (%) (↑) on SPATIALAB-OPEN by Question Categories.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline							
	73.53	50.19	70.30	69.81	65.48	62.87	64.93

manages only 30.50% despite its considerable size. Category-wise, Orientation and 3D Geometry are where the strongest models shine (several exceed 60% on either), whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are far more variable and consistently harder across the board. Tellingly, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) land only in the low-to-mid 40s overall, so specialization plainly does not translate into higher MCQ accuracy on our diverse question set. Depth & Occlusion and Size & Scale split the field, some architectures handle occlusion well while others fail outright, revealing sharply differing inductive biases or training data. These patterns show decisively that instruction-tuning and reasoning capability boost geometric and orientation reasoning, while multi-step grounding tasks such as navigation remain a clear, unresolved weakness.

SPATIALAB-OPEN. Open-ended scores in Table 3 fall far below the MCQ results and spread widely: overall accuracy ranges from about 9.6% (InternVL3.5-1B) to just 40.9% (GPT-5-mini) against a human baseline of 64.9%, leaving a large gap to close. Proprietary models lead the leaderboard on open-ended tasks (e.g., GPT-5-mini 40.93%, GPT-4o-mini 26.00%, Gemini-2.5-Flash 30.93%), and reasoning-tuned variants stand out unmistakably: o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) outperform many open-source counterparts, proving that instruction/CoT-style tuning substantially improves generative outputs. Most small open-source models cluster at the bottom (many in the teens; InternVL3.5-1B = 9.64%, Qwen-VL2.5-3B = 12.93%), while even the largest open models climb only to the low-20s (e.g., InternVL3.5-72B = 23.36%, Qwen-VL2.5-72B = 24.64%), confirming that scale alone buys only modest gains. Spatial-specialist models (SpaceOm, SpaceThinker, SpaceQwen) reach a mere 10–13% overall, establishing that architectural or task specialization does not deliver robust open-ended generation on our diverse question set. Across subtasks, top models fare relatively better on orientation and certain relative-position items (e.g., GPT-5-mini Rel.Pos = 49.53%, Ori = 37.13%), whereas depth & occlusion, size & scale, and above all spatial navigation stay consistently hard (many models score <30% on these subtasks). This pattern confirms our earlier finding: multi-step grounding and perceptual reasoning are the decisive bottlenecks for open generative answers.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

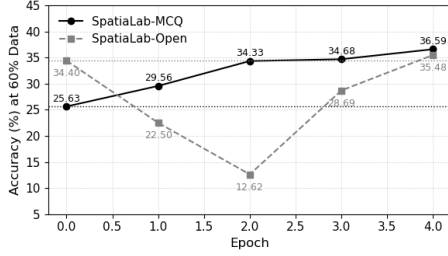
Across 25 models, the average MCQ→open-ended performance gap in SPATIALAB is a substantial 23.0% ($\sigma = 5.5\%$), with subtask means ranging from 22.89% (spatial navigation) to 24.57% (3D geometry). Specialist spatial reasoning models suffer the largest gaps (about 27%), driven by spatial navigation (up to 36.68%) and orientation (34.44%), whereas reasoning-oriented models hold much smaller gaps (around 19%) and lower subtask variance, a direct consequence of the stabilizing effect of instruction-tuning and CoT decoding. Correlation analysis pinpoints spatial navigation as the dominant source of overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also contributing. Negative or near-zero gaps (e.g., Llama-3.2-11B: -1.98 in depth & occlusion; o4-mini: -3.18 in relative position) expose how poorly designed MCQ distractors misrepresent true competence. We attribute this format sensitivity to four causes: MCQ’s structural advantage, specialization bias toward categorical tasks, the sequential-reasoning demands of spatio-navigation, and calibration differences in generative output. The conclusion is unavoidable: MCQ alone overestimates practical spatial reasoning ability. We therefore recommend complementing MCQ with open-ended evaluation, auditing distractors, employing stepwise generation or instruction-tuning, and reporting per-subtask diagnostics to recover true model performance and eliminate format-dependent artifacts. More details are in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. Our analysis (Appendix E) establishes that VLMs achieve selective peaks but lack holistic spatial competence. Closed-source models reach the highest scores (e.g., 85.71% in *Stacking Orientation*), yet collapse to near-chance in *Relative Size Comparison*. Open-source scaling raises the ceiling (80.56% in *Corner/Angle Positioning*) but never prevents catastrophic failures (2.0% in *Object Rotation*). Reasoning-augmented and spatially specialized models contribute localized gains, in occlusion inference or navigation, yet stay capped below 55% on many physical-abstraction tasks (*Gravity Effects*, *Stability Prediction*). Error distributions expose systematic weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*), while the comparatively strong *Obstacle Avoidance* results betray shortcut exploitation rather than robust planning. Taken together, these patterns prove that current VLMs lean heavily on surface-level correlations and lack stable encodings of orientation, physics, and compositional logic. Progress will require both richer training distributions (e.g., embodied and physics-driven data) and architectural mechanisms for geometric and reference-frame grounding.

SPATIALAB-OPEN. Error analysis of SPATIALAB-OPEN (Appendix F) shows that even top-performing closed-source models (e.g., GPT-5-mini: 58.14% in *Directional Relations*) collapse on tasks like *Proximity Gradients*, proving that their dominance is strictly task-specific. Large open-source models achieve strong results in *Depth & Occlusion* (e.g., 60.0%) yet still fall to 9.3% on *Proximity*, confirming that scale alone cannot buy robustness. Smaller models display isolated strengths (e.g., 26.0% in *Relative Size Comparison*) but break down entirely elsewhere (e.g., 0.0% in *Betweenness*), exposing the hard limits of compact architectures. Reasoning-tuned systems are promising, Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness*, yet peer models drop to single digits, revealing fragile integration. Specialized spatial models perform worst of all, rarely exceeding 20% and frequently failing core relational tasks. These failure patterns lay bare systemic weaknesses in occlusion handling, orientation, and multi-step relational chaining, while partial success on size-scale cues betrays reliance on superficial correlations. Three root causes emerge unambiguously: missing representations for spatial analogies, architectural limits tied to scale and training diversity, and brittle reasoning pipelines, together mandating unified geometric encodings, physics-aware reasoning, and embodied data to move from isolated peaks to consistent spatial competence.

Qualitative Error Analysis. To pinpoint model weaknesses, we conducted a qualitative error analysis (see Appendix I) and found that failures cluster into a small number of recurring classes rather than scattering as random noise. The dominant errors are spatial mislocalization, where models confuse referents in crowded scenes; perspective and scale mistakes, where object-size priors override image-based cues; and occlusion or ordering failures, especially with thin or partially hidden structures. We also observe attribute confusion, mixing perceptual properties with functional ones, and open-ended rationalizations that produce fluent but visually ungrounded narratives. These errors recur consistently across architectures and prompting styles, evidence of structural biases in current



(a) SFT performance trends over epochs.

Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(b) Accuracy before vs. after SFT across categories, showing decisive MCQ gains alongside flat-to-negative open-ended transfer.

Figure 4: SFT training results: (a) learning trends over epochs, and (b) final accuracy values, confirming that SFT lifts MCQ accuracy while failing to transfer to open-ended reasoning.

VLMs rather than dataset-specific noise. Crucially, failure rates spike when multiple cues must be fused, for example combining depth ordering with relative size, and confidence calibration is especially poor in open-ended settings. Our diagnostic protocols confirm that models routinely ignore minimal but decisive visual features, rely on brittle heuristics, and fail to update when counterfactual edits are applied. The root causes are clear: insufficient object-centric binding, absent geometric supervision, and training objectives that reward plausibility over grounding. Together, these findings establish that current VLMs achieve strong coarse perception yet fail at multi-cue integration and grounded reasoning, making the case for geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Reasoning-enabled models consistently outperform their baselines in both MCQ and open-ended formats, with the largest gains in relational and orientation tasks (e.g., +13.1% in Relative Positioning for MCQ). Reasoning clearly stabilizes open-ended performance, but its gains remain uneven across categories (notably declining in Size & Scale), demonstrating that reasoning modules sharpen logical consistency yet leave grounding and scale sensitivity unsolved. Detailed per-category comparisons appear in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting delivers little benefit and frequently reduces accuracy (Table 19), with orientation the sole category showing consistent gains. Step-by-step reasoning thus helps with directional alignment but fails outright on tasks demanding robust perceptual grounding, depth, scale, and spatial navigation. Unlike in textual reasoning, CoT here amplifies flawed priors rather than correcting them, making it clear that solving SPATIALAB requires perceptual understanding, not merely logical chaining (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Adding self-reflection to CoT prompting, where models are explicitly prompted to review their prior reasoning step-by-step and correct mistakes (see Figure 8 in Appendix for the prompt), yields only modest benefits in multiple-choice settings, chiefly for geometry and depth, and fails to generalize to open-ended tasks (Appendix H.3). This confirms decisively that reflective reasoning cannot compensate for missing perceptual grounding, a core limitation of current VLMs.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning (SFT) on Qwen2.5-VL-3B-Instruct using a stratified 40% split of SPATIALAB (~560 samples) consistently improves MCQ accuracy across every spatial reasoning category, yet delivers little, and sometimes negative, transfer to open-ended tasks. This divergence is a clear signature of overfitting to task-specific answer distributions, confirming that SFT *teaches to the test* rather than fostering generalizable reasoning (Wang et al., 2025). The sharp drop in open-ended performance exposes biased internal representations, tuned for categorical discrimination but unstable in generative settings. The likely mechanism is catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that reward discrete-choice alignment over spatial reasoning in language. In short, SFT boosts performance in structured formats but risks an algorithmic straight-jacket, lifting accuracy on constrained tasks while stalling progress in naturalistic spatial reasoning. More details are available in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To probe agentic capabilities, we employ SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) to

perform structured spatial reasoning on images. The framework decomposes complex visual reasoning into specialized sub-tasks handled by dedicated agents, with modules for object segmentation, attribute extraction, spatial relation mapping, and transformation tracking. By consolidating these outputs into a unified structured representation, the system propagates low-level perceptual cues through higher-order relational inferences. We implemented the pipeline with Gemini-2.5-Flash at low temperature to guarantee deterministic outputs across the multi-stage refinement process. The results are striking in the Orientation category, improving by +8.00% in MCQ and a remarkable +36.00% in open-ended evaluation. But the gains are far from universal: Depth & Occlusion and Spatial Navigation suffered severe declines of -24.00% and -12.00% in open evaluation, respectively. These drops make the mechanism explicit, when inherent perceptual priors such as depth or occlusion cues are weak, multi-step reasoning reinforces misconceptions instead of correcting them. The verdict is clear: agentic workflows sharpen tasks reducible to sequential alignment but cannot substitute for fundamental perceptual grounding in complex 3D environments. More details are available in Appendix H.5.

6 DISCUSSION

Spatial reasoning is central to embodied intelligence, yet most benchmarks rely on synthetic scenes, templated questions, or narrow task scopes that ignore real-world visual variability. We close this gap with SPATIALAB, a benchmark of 1,400 visual question-answer pairs covering six major spatial categories and 30 subcategories, including 3D geometry, occlusion, and navigation. The benchmark deploys both Multiple-Choice and Open-Ended formats to cleanly separate option-based pattern matching from genuine generative reasoning. A multi-stage review process and complexity checks spanning lighting, texture, and spatial diversity guarantee its quality. Evaluations of more than 25 contemporary models against human baselines establish SPATIALAB as a precise diagnostic instrument for exposing the weaknesses of current vision-language systems.

Detailed error analysis uncovers systematic failures in handling transparency, multi-step navigation, and occlusion, demonstrating that models lean on surface-level texture cues rather than internalized 3D geometric representations. Supervised Fine-Tuning (SFT) yielded clear gains on structured MCQ tasks yet simultaneously triggered instability in open-ended reasoning, a textbook case of catastrophic forgetting of linguistic priors when optimizing for discrete answers. Chain-of-Thought proved effective for sequential tasks like Orientation but failed to correct fundamental perceptual errors in Depth and Scale, amplifying hallucinations rather than resolving them. Agentic decompositions exhibited the same trade-off, improving narrow alignment tasks while degrading holistic scene understanding. These divergent behaviors settle the matter: scaling laws and reasoning scaffolds alone are insufficient, and achieving human-level spatial intelligence will require architectural innovations that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

Spatial reasoning remains a core bottleneck for VLMs, above all in visually complex, real-world settings. SPATIALAB contributes 1,400 diverse QA pairs across six spatial categories and thirty subcategories, enabling fine-grained evaluation through both MCQ and open-ended formats. Even after applying several performance-improving methods, our experiments demonstrate substantial, persistent gaps between state-of-the-art models and human performance, with open-ended generation dropping a further 10–25% relative to multiple-choice accuracy. Error analysis traces these gaps directly to depth, occlusion, 3D geometry, and multi-step navigation, all symptoms of reliance on superficial cues rather than grounded spatial representations. SPATIALAB exposes these limitations and delivers a structured testbed for diagnosing spatial competence. Our benchmark enables holistic evaluation and yields actionable insights for building VLMs with human-aligned spatial understanding. Future progress will demand richer training signals (e.g., embodied interaction, action-conditioned feedback) and architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We have taken multiple steps to ensure ethical compliance throughout the development of this work. All metadata associated with the images was removed to protect privacy, and any personally identifiable features (such as human faces or license plates) were minimized or excluded wherever possible. The dataset was carefully curated to avoid copyright concerns, with diligent attention paid to sourcing and licensing, though minor oversights may occur despite best efforts. The benchmark construction process was designed with transparency and fairness in mind, ensuring that no demographic groups were singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To facilitate reproducibility, we provide a detailed description of the dataset construction process, annotation pipeline, and evaluation protocol in the main text and appendix. All filtering, quality-control, and validation steps are explicitly documented, with clear definitions of task categories and annotation criteria. Hyperparameters, model configurations, and training procedures are reported in the appendix to support consistent replication of our results.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning, the capacity to understand geometric structures and relational layouts in visual scenes, is central to embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) tested compositional reasoning with synthetic or curated templates; these enabled controlled analysis but offered scant ecological validity. With the advent of large vision-language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023), spatial reasoning has been revisited on more naturalistic datasets, yet systematic weaknesses persist. Despite strong performance on captioning and visual QA, these models repeatedly fail on tasks requiring depth ordering, occlusion inference, or consistent reference-frame alignment. This discrepancy exposes a broader truth: scaling alone does not guarantee progress in structured spatial understanding.

Spatial reasoning benchmarks. A range of benchmarks have recently been proposed to probe these gaps (Table 1). SCANQA (Azuma et al., 2022) was among the first embodied QA datasets, requiring reasoning about 3D indoor environments, but its limited scale (800 scenes) and shallow complexity sharply restrict generalization. VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) extended scope to real-world or household contexts yet leaned on template-driven questions, which suppress linguistic variability and overrepresent simple binary or multiple-choice relationships. More recent efforts, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), broadened task variety but remain constrained by scale (2.2K and 211 samples, respectively) and by template-heavy or narrowly manual annotation. SPATIALRGPT-BENCH and BLINK-SPATIAL introduced mixed domains and up to 14 categories, improving diversity, yet still confine themselves to static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. The trend toward embodiment is visible in datasets like ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes with robot-centric questions. Yet its template-based annotation renders questions unnatural and readily gamed with heuristic shortcuts. SPATIALVLM (Chen et al., 2024b) offers unprecedented scale (billions of QA pairs), but its two-category coverage disqualifies it for fine-grained diagnostic evaluation. At the opposite extreme, high-quality manual datasets such as VSI-BENCH (Yang et al., 2024) achieve superior annotation fidelity, but with only 288 scenes they are far too small for training or robust cross-model comparison. In complexity, most existing benchmarks cap at “medium” (Mm in Table 1), rarely reaching the “hard” or “extremely hard” levels that demand multi-step causal inference. This leaves a wide, unaddressed evaluation gap for next-generation models.

Evaluation protocols. Evaluation type and analysis depth matter just as much. Most prior work employs close-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which simplify scoring but throttle the expressivity of model outputs. A few datasets such as ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN introduce open-ended evaluation, but only SPATIALAB-OPEN pairs it with extremely high complexity, manual puzzle-style annotations, and comprehensive error analysis. This combination unlocks a deeper understanding of systematic weaknesses beyond accuracy, including calibration and cross-model divergence, which existing datasets rarely capture. Absent this granularity, reported gains are misleading, since models may simply overfit dataset-specific biases rather than acquire robust spatial competence.

Positioning of SPATIALAB. SPATIALAB directly resolves these limitations. Where prior datasets span 2 to 14 task categories, SPATIALAB covers 30 distinct categories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation, delivering unprecedented task diversity. Unlike template-based benchmarks, every question is manually authored and puzzle-inspired, guaranteeing linguistic richness and shutting down shortcut opportunities. SPATIALAB is moreover fully embodied and multi-modal, encompassing both static and dynamic visual contexts, and thereby bridges robotics-style datasets (ROBOSPATIAL) and web-scale corpora (SPATIALVLM). As Table 1 shows, even the strongest baselines, InternVL3.5-72B and GPT-5-mini, remain well below 60%, and the open-ended variant yields a mere 40.93% accuracy. This large gap proves both the difficulty of the benchmark and its power to drive future research.

Toward next-generation evaluation. By combining high difficulty, broad coverage, manual annotation, and open-ended tasks, SPATIALAB advances the landscape of spatial reasoning benchmarks from template-driven, narrow evaluations to comprehensive, embodied testing. This design not only assesses whether a model answers correctly but also identifies systematic error patterns, probes causal and counterfactual reasoning, and measures calibration reliability. SPATIALAB thus provides a rigorous foundation for the next phase of multimodal research, complementing datasets like MIND THE GAP and VLM4D that explore temporal or generative aspects but remain narrower in coverage. We expect SPATIALAB to serve as the high-bar diagnostic tool that pushes models beyond surface-level recognition toward genuine grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB within the broader landscape, we conducted a detailed examination of existing spatial reasoning benchmarks. Earlier efforts laid important foundations, but each carries structural and methodological constraints that sharply limit its value for evaluating current VLMs. These constraints fall into three areas: synthetic-domain bias, limited task scope, and restrictive evaluation formats. Table 4 summarizes the major limitations we identify.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA rely on synthetic environments or template-generated linguistic structures. These settings offer controlled conditions for isolating compositional reasoning, but the visual domain lacks real-world complexity and the linguistic space is narrow, so models exploit template regularities instead of grounded spatial interpretation. Related simulated benchmarks such as ROBOSPATIAL inherit the same flaw, producing a large sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but remain confined to indoor scenes, precluding assessment in the outdoor or mixed environments essential for embodied agents. High-quality manual datasets such as VSI-Bench offer richer annotation but are too small in scale for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but lack detailed taxonomies, rendering them unusable for diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, lean heavily on Binary or MCQ formats that inflate performance by enabling option elimination rather than requiring generative spatial reasoning. SPATIALAB eliminates these issues by combining MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations we identify in representative prior spatial benchmarks, set against SPATIALAB. Limitations are organized by domain constraints, evaluation format, and task taxonomy.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale ($< 1K$ scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~ 200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning demands far more than simple object recognition; it hinges on how objects relate, align, and interact in physical space. We introduce *SpatialLab*, a benchmark that rigorously tests fine-grained spatial understanding across diverse categories. The tasks in SpatialLab mirror how humans naturally describe and reason about environments, spanning relative positioning (e.g., *What’s to the left of the microwave?*) through geometric inference (e.g., *Would this stack fall if another book is added?*). Each category decomposes into subcategories that isolate subtle yet essential dimensions of reasoning, including occlusion, orientation, navigation, and physical effects such as gravity. By framing every subtask in accessible, everyday terms, SpatialLab forces models beyond

object detection and into structured spatial interpretation. This design directly probes how well systems generalize across the contexts that matter for robotics, AR/VR, autonomous driving, and human-computer interaction. The following sections give concise, task-level descriptions for every category and subcategory in SpatiaLab.

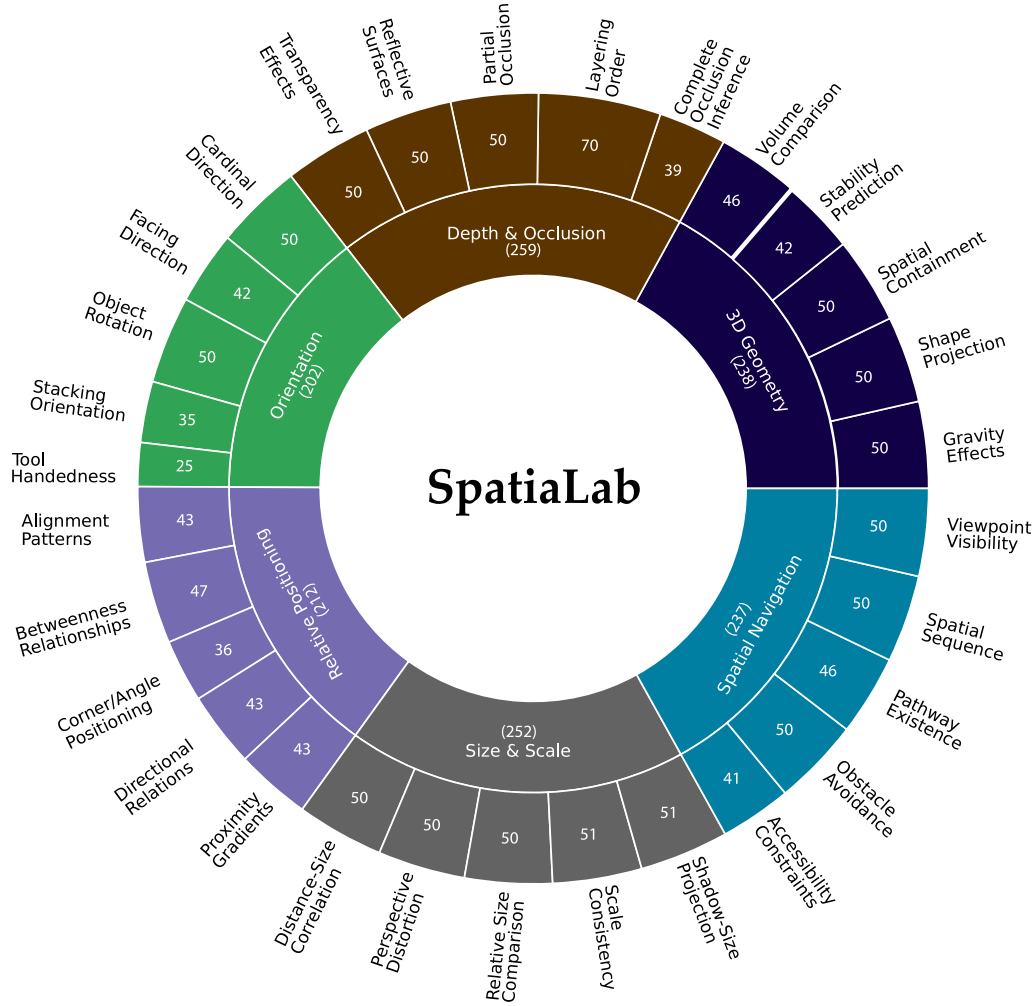


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each category decomposes into five subcategories, yielding a comprehensive thirty task types in total.

The following details the rationale and real-world utility driving each task design, per category:

B.1 RELATIVE POSITIONING

Directional Relations. Directional relations describe how objects are positioned relative to one another, such as left, right, above, or below. Mastering these relations lets models interpret spatial layouts as naturally as humans describe scenes. For example, identifying what lies immediately to the left of an appliance is essential for scene understanding and robot navigation.

Proximity Gradients. Proximity reasoning identifies which objects are closest or farthest from a reference point. This is central to questions like “Which object is nearest to the doorway?” Models that estimate gradients of closeness directly enable navigation tasks and assistive technologies.

Alignment Patterns. Alignment concerns whether objects follow a specific orientation or order, such as being stacked vertically or arranged in a straight line. Recognizing alignment patterns is key to

1242 understanding organization in environments like shelves, tables, or warehouses, and is critical for
1243 robotic sorting and automated inspection.

1244 **Betweenness Relationships.** Betweenness identifies objects that occupy the middle ground between
1245 two references. Asking what sits between the lamp and the armchair, for instance, demands reasoning
1246 about spatial continuity. This drives navigation, retrieval, and situational awareness tasks.

1247 **Corner/Angle Positioning.** Corners and angled positions serve as reference anchors in spatial
1248 reasoning. Questions such as “What’s positioned in the back-right corner?” force the model to
1249 localize objects within constrained regions. This reasoning is vital in robotics for mapping and
1250 placement.

1251 1252 1253 B.2 DEPTH & OCCLUSION 1254

1255 **Layering Order.** Layering requires distinguishing which objects appear in front versus behind.
1256 Identifying the frontmost item on a table, for example, directly reflects depth reasoning. This is
1257 especially valuable in AR/VR applications where visual stacking shapes user experience.

1258 **Partial Occlusion.** Partial occlusion occurs when one object hides part of another from view.
1259 Recognizing the hidden item behind a coffee cup, for instance, demands filling in missing information.
1260 This hardens perception systems operating in cluttered environments.

1261 **Complete Occlusion Inference.** Objects are sometimes fully hidden, yet clues like shadows or
1262 bulges reveal their presence. Inferring what lies under a blanket forces the model to exploit indirect
1263 evidence, mirroring human inference under uncertain visual conditions.

1264 **Transparency Effects.** Transparent surfaces, such as glass or water, allow partial visibility of objects
1265 behind them. Determining what’s visible through a fish tank demands blending occlusion reasoning
1266 with transparency perception, a capability central to simulation, graphics, and embodied AI.

1267 **Reflective Surfaces.** Reflective reasoning requires models to interpret mirrors, shiny metals, or
1268 glass reflections. Identifying what is reflected in a mirror involves indirect spatial mapping. This is
1269 essential for safety in autonomous vehicles and situational awareness in robotics.

1270 1271 1272 B.3 ORIENTATION 1273

1274 **Cardinal Direction.** Cardinal reasoning concerns compass directions like north, south, east, and
1275 west. Determining which way a front door faces requires integrating global orientation cues. This is
1276 critical for navigation and geospatial alignment.

1277 **Object Rotation.** Rotation reasoning measures how much an object has been turned around an axis.
1278 Estimating that a chair is rotated by some degrees is crucial for accurate 3D modeling and lets robots
1279 align objects correctly for manipulation.

1280 **Facing Direction.** Facing direction specifies the orientation of an object’s front side. Determining
1281 whether a monitor faces north, for example, requires mapping its surface orientation to global
1282 references, a capability important in surveillance and robotics.

1283 **Stacking Orientation.** Stacking reasoning determines whether objects are piled vertically, horizon-
1284 tally, or in mixed orientations. Recognizing such layouts ensures stability in warehouses and homes
1285 and is critical for safe robotic stacking.

1286 **Tool Handedness.** Certain tools, like scissors or gloves, are designed for left- or right-handed use.
1287 Identifying handedness demands fine-grained shape and orientation perception, directly enhancing
1288 assistive robotics in human-centered environments.

1289 1290 1291 B.4 SIZE & SCALE 1292

1293 **Relative Size Comparison.** Relative comparisons assess whether one object is larger, taller, or
1294 smaller than another. Judging if a vase is taller than a statue demands proportional reasoning, which
1295 is crucial for packing, fitting, and recognition tasks.

Perspective Distortion. Perspective makes distant objects appear smaller than nearby ones even when the actual size differs. Resolving illusions like a faraway car looking smaller than a nearby bicycle requires compensating for depth cues, making models robust to camera perspectives.

Distance-Size Correlation. Here, size judgments depend on distance estimation. A nearby bicycle appearing larger than a far car must not confuse true size reasoning. This is fundamental to navigation, AR/VR, and visual grounding.

Scale Consistency. Scale reasoning checks whether object sizes are plausible in context. A mouse larger than a chair signals inconsistency that models must detect, strengthening realism in simulated environments.

Shadow-Size Projection. Shadows provide indirect cues about object size and light source position. Estimating how tall a light must be from shadow length demands geometric reasoning, which is valuable in graphics, architecture, and physics-based modeling.

B.5 SPATIAL NAVIGATION

Pathway Existence. Pathway reasoning determines whether a clear route exists between two points. Deciding if one can reach the desk without moving chairs, for example, tests navigability, an ability that underpins indoor robot planning.

Obstacle Avoidance. Avoidance requires finding routes that minimize collisions. Identifying the clearest path from door to window maps directly to robotic and human navigation, and is essential in autonomous driving and indoor mobility aids.

Viewpoint Visibility. From a given viewpoint, certain objects are blocked while others remain visible. Determining what is obscured when sitting on the couch, for example, requires simulating perspective, strengthening AR/VR realism and robotic vision.

Spatial Sequence. Sequence reasoning identifies the order of encounters along a path. Walking from the kitchen to the balcony, one passes the dining table first. Such reasoning replicates human navigation memory.

Accessibility Constraints. Constraints identify which objects are blocked from access by obstacles. A cabinet may be unreachable because of a table in front, for instance. This is crucial for assistive AI and ergonomic design.

B.6 3D GEOMETRY

Volume Comparison. Volume comparison determines which object occupies more three-dimensional space. Judging whether a box is larger than a sphere drives packing and containment tasks, and is important for logistics and simulation.

Stability Prediction. Stability reasoning predicts whether adding or moving objects will topple a stack. Deciding whether placing a book will collapse a pile mirrors human intuition and supports safety in robotics and construction.

Shape Projection. Projection tasks determine what shape an object casts under given rotations or lighting. A sphere always projects a circle, while a cube may project a square or hexagon. Such reasoning strengthens geometric understanding.

Spatial Containment. Containment determines whether one object fits inside another. Reasoning about whether a ball fits in a box supports packing and design, and aligns with intuitive geometry.

Gravity Effects. Gravity reasoning predicts motion outcomes under physical forces. A rolling ball may stop under a table due to friction and barriers. This strengthens real-world simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

We recruited annotators aged between 24 and 30 years, all of whom were engineering graduates. They participated voluntarily and worked independently without external incentives. Before annotation began, we delivered structured training sessions that established clear task requirements and enforced consistency across annotators. We upheld all ethical and quality control standards throughout the process. This approach guarantees that the resulting annotations are both reliable and tightly aligned with the dataset design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Before annotation began, we designed a structured training program that enforces consistency and reliability across the full taxonomy of tasks. We first introduced annotators to the six core categories (*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*) and their thirty subcategories. For each subcategory, we supplied curated examples that illustrate the corresponding spatial relations (e.g., directional relations, occlusion inference, stability prediction) together with counter-examples that expose common pitfalls. Beyond the taxonomy, we trained annotators to account for six meta-dimensions of image variability: lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints, ensuring that questions explicitly exploit visual diversity. We demonstrated how these factors interact in real-world settings, such as reasoning about occlusion under low-contrast lighting or maintaining scale consistency under perspective distortion.

We also introduced the two evaluation modes: multiple-choice and open-ended. In multiple-choice settings, annotators learned to design precise distractors that test fine-grained discrimination, while in open-ended settings they practiced formulating prompts that demand compositional reasoning. To ground this training, we ran calibration sessions in which annotators generated both types of questions for sample images and iteratively refined their designs with feedback. To situate task design within the broader research landscape, we reviewed prior benchmarks (e.g., SpatialBot-Bench, EmbSpatial, SpatialMM) with annotators and dissected their limitations. These resources emphasize simplified relations, synthetic or staged scenes, and narrow reasoning types, while neglecting natural complexity such as shadows, reflections, or gravity-sensitive stability. This gap analysis establishes the need for richer, more ambiguous, and dynamically layered spatial contexts.

Finally, we provided detailed demonstrations of the annotation protocol. These demonstrations specified how to select images with sufficient spatial complexity, how to map questions unambiguously to subcategories, and how to avoid trivial formulations. We directed annotators to prioritize tasks requiring multi-step reasoning, such as depth layering, relational chains, or cross-view scale inference, over binary labels. Through this systematic training and iterative feedback, the final dataset achieves a rigorous balance of taxonomic coverage and real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

The prompts used for the answer model are shown in Figure 6. We set `temperature` to 0 and `max_tokens` to 10, while keeping all other generation parameters at their default values.

D.2 OPEN-ENDED EVALUATION

The prompts used for both the answer model and the judge model are provided in Figure 6. We set `temperature` to 0.7 and `max_tokens` to 1200, with all other parameters kept at their default values.



Figure 6: Prompts used throughout evaluation.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To evaluate *Gemini-2.5-Flash* as an automatic judge, we assembled a balanced set of 240 samples spanning all categories. Each sample included the full evaluation setup (question, ground truth, and model output), which three human annotators scored independently. Annotators assigned binary scores: 1 for correct and 0 for incorrect. We measured the LLM’s judgments against the human annotations using multiple complementary agreement metrics, computed both overall and per category. Specifically, we employed **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, defined as follows.

Cohen’s Kappa. Cohen’s kappa (κ) quantifies pairwise agreement between two raters while correcting for chance agreement. For a binary task, it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o is the observed agreement and p_e is the expected agreement by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here, $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c . An $\kappa = 1$: perfect agreement; $\kappa = 0$: agreement equals chance; $\kappa < 0$: systematic disagreement) We computed Cohen’s kappa between the LLM and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), and between the LLM and the majority vote of the annotators (κ_{majority}).

Accuracy with Majority Vote. We further measured raw agreement between the LLM and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the LLM’s decision for sample i , and y_i^{maj} is the majority vote among annotators. Unlike Cohen’s kappa, accuracy does not correct for chance agreement; it simply reflects the proportion of identical labels.

Fleiss’ Kappa. To quantify inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to multiple raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is

$\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then computed as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Higher values signal stronger inter-annotator reliability beyond chance.

Table 5: Agreement metrics between Gemini-2.5-Flash as an automatic judge and three human annotators across all categories. The table reports Cohen’s kappa with the majority vote and each annotator, raw accuracy against the majority, and Fleiss’ kappa among all annotators. Bold values mark the highest agreement within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 establishes that the LLM judge (Gemini-2.5-Flash) attains strong overall agreement with human annotators, achieving a Cohen’s kappa of 0.738 with the majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority reaches 0.880 overall, while Fleiss’ kappa among human annotators is 0.774, confirming substantial consistency. Several categories, such as 3D Geometry, reach perfect agreement with $\kappa = 1.000$ and accuracy of 1.000. Depth & Occlusion likewise achieves high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) show moderate agreement yet remain within acceptable reliability ranges.

These results establish that Gemini-2.5-Flash matches human evaluators in scoring correctness, with Cohen’s kappa values predominantly in the substantial range (0.61–0.80) and several categories reaching almost perfect agreement (higher than 0.81). Fleiss’ kappa of 0.774 confirms that the human annotators themselves are highly consistent, validating the reliability of the reference scores. The LLM achieves perfect or near-perfect agreement in clearly defined categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933), demonstrating accurate pattern recognition. Moderate agreement in categories like Orientation and Size & Scale reflects task complexity and inherent subjectivity rather than model shortcomings. Overall, these findings demonstrate that the LLM-as-judge delivers robust reliability and serves as a practical automated scoring tool that closely mirrors human evaluation.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) deliver the strongest results across the benchmark. **GPT-4o-mini** sets the overall ceiling, reaching 85.71% in *Stacking Orientation*, the highest score anywhere in the suite. **Gemini-2.0-Flash** dominates occlusion and projection, achieving 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*, and **GPT-5-mini** scores 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. Yet decisive weaknesses persist: **Claude 3.5 Haiku** collapses to 32.0% in *Relative Size Comparison*. On average, closed-source systems outperform open-source ones by 15–25 percentage points across several categories, but within-model variation is severe, routinely exceeding 50 percentage points. This establishes that closed-source training pipelines raise the ceiling without solving the challenge of uniform spatial competence.

Open-source Large Models. Scaling to large open-source models (Table 7) delivers clear improvements yet exposes severe fragilities. **InternVL3.5-72B** achieves 80.56% in *Corner/Angle Positioning*, one of the highest open-source results, and both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** reach 66.67% in *Complete Occlusion Inference*. These peaks coexist with outright failures: **llama-3.2-11b-vision-instruct** scores just 2.0% in *Object Rotation*. **llama-3.2-90b-vision-instruct** also climbs to 71.43% in *Stacking Orientation*, but only by trading away other categories.

Table 6: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

The within-model spread routinely exceeds 70 percentage points, a clear signature of brittle specialization, and gravity reasoning stalls at 56.0%. The evidence establishes that scaling raises peaks without resolving structural weaknesses: size amplifies selective strengths while leaving foundational representation gaps intact.

Open-source Small Models. Open-source small models (Table 8) exhibit sharp, decisive contrasts across sub-categories. **qwen-2.5-vl-7b-instruct** reaches 74.29% in *Stacking Orientation*, while **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. *Pathway Existence* stays low, with the best models tied at 34.78%, and gravity-related reasoning is equally weak, capping at 50.0% for **InternVL3.5-4B** and **Gemma-3-4B-it**. The within-model spread routinely exceeds 60 percentage points, exposing brittle generalization. These patterns prove that smaller models are cue-dependent: they succeed where visual templates are strong and fail wherever the task demands deeper spatial abstraction. This inconsistency demonstrates that representation learning is uneven across sub-categories. Small models thus deliver localized competence but lack robustness across the full spectrum of spatial reasoning.

Table 8: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Table 9: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Reasoning Models. Reasoning-enhanced models (Table 9) deliver notable gains on select tasks yet remain decidedly inconsistent overall. **o4-mini** reaches 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, outperforming most open baselines, and **Gemini-2.0-Flash-Thinking** achieves 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*, evidencing stronger geometric reasoning. The strongest results belong to **Gemini-2.5-Flash-Thinking**, which scores 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. The same family nonetheless fails elsewhere, dropping to 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. This variance proves that reasoning mechanisms deepen performance in certain sub-categories without broadening coverage, with within-model spreads frequently exceeding 35 percentage points. Reasoning augmentation therefore adds localized capacity while leaving structural gaps in spatial understanding intact: these models establish promise, not generalizable competence.

Table 10: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

Spatial Reasoning Models. Spatially specialized models (Table 10) deliver targeted improvements yet lack generality. **SpaceQwen2.5-VL-3B-Instruct** achieves 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, decisively surpassing many open-source baselines in size-scale reasoning, and leads in navigation with 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*, as expected for domain-specific reasoning. Performance nonetheless collapses on other tasks, falling to 24.0% for *Spatial Containment*, and these models rarely exceed 55.0% even in their strongest areas, with sub-category spreads frequently surpassing 30 percentage points. The evidence establishes that specialization drives narrow peaks but does not produce holistic competence: domain targeting sharpens select abilities while failing to integrate them into a robust, general framework.

Overall Discussion. Across all categories (Table 8, Table 7, Table 6, Table 10, Table 9), the fragmentation is consistent and unmistakable. Small open-source models peak at 74.29% in *Stacking Orientation* yet collapse to 12.0% in *Cardinal Direction*. Larger models raise ceilings to 80.56% in *Corner/Angle Positioning* but still fail catastrophically at 2.0% in *Object Rotation*. Closed-source systems post the strongest scores, with **GPT-4o-mini** reaching 85.71% in *Stacking Orientation*, yet fall to 32.0% in *Relative Size Comparison*. Specialized models excel in navigation (53.66%) and stability (50.0%) but stay capped below 55% overall, and reasoning models climb to 71.79% in *Complete Occlusion Inference* while others remain stuck at 34.0%. Within-model spreads routinely exceed 50 percentage points, exposing brittle internal consistency, and sub-categories requiring physical abstraction, *Gravity Effects*, *Stability Prediction*, *Spatial Containment*, consistently score below 35–40% regardless of category. These failures trace to two root causes: the absence of stable encodings of orientation and physics, and an over-reliance on surface-level correlations. The variance from 85.71% highs to 2.0% lows proves that models pattern-match rather than reason robustly about space. Scaling, proprietary training, and reasoning augmentations each confer benefits, but none produces general coverage; each yields isolated peaks that coexist with glaring blind spots. The collective picture is fragmented competence, not integrated ability. Progress in spatial reasoning therefore demands architectural changes that embed consistent geometric grounding and physics-

Table 11: Distribution of question counts by the number of models (0–5) that correctly answered each item across categories and sub-categories in SPATIALAB-MCQ. “0” indicates no model answered correctly, while “5” indicates only five models succeeded to answer these among all models tested.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

aware mechanisms. In sum, our results establish both remarkable selective gains and the persistence of fundamental gaps in holistic spatial intelligence.

E.2 MODEL FAILURE ANALYSIS

The distribution across categories and sub-categories in Table 11 exposes several decisive patterns in model performance. Categories such as 3D Geometry and Size & Scale show relatively balanced distributions, yet both retain a substantial proportion of failures where no model answered correctly, proving that even basic geometric invariances are not reliably internalized. Depth & Occlusion, by contrast, exhibits a heavier tail in the mid-range (2–4 correct), evidencing partial but inconsistent reasoning about layered scenes. Orientation stands out as the weakest category overall, with especially low totals in Stacking Orientation and Tool Handedness, which demonstrates that reasoning about

subtle directional or embodied affordances is poorly captured. Relative Positioning is mixed: some tasks like Proximity Gradients show moderate success, while others such as Corner/Angle Positioning have many near-zero correct counts, exposing brittleness in compositional reasoning over reference frames. Spatial Navigation performs comparatively stronger, particularly in Obstacle Avoidance and Viewpoint Visibility, but still incurs significant errors in Pathway Existence, where relational chaining is required. Aggregated, no category achieves uniformly high accuracy, and the Grand Total spreads evenly across the 0–5 bins, proving that success is task-dependent rather than reflecting general spatial competence. Models thus display isolated strengths but lack systematic reliability across spatial dimensions.

These results trace current vision–language failures to several root causes rooted in representational and reasoning bottlenecks. High error rates in Stacking Orientation and Tool Handedness expose limited grounding in physics-based and embodied reasoning, consistent with models trained primarily on static internet imagery rather than interactive or task-driven data. The difficulty in Corner/Angle Positioning and Pathway Existence reflects inadequate multi-step relational chaining and confirms weaknesses in handling recursive spatial logic. The inconsistency in Transparency Effects and Reflective Surfaces underscores the challenge of visual phenomena that require non-local cues, such as secondary reflections or occluded geometry. Comparatively better performance in Obstacle Avoidance, by contrast, indicates that models exploit dataset-bias shortcuts, such as frequent co-occurrence patterns of walkable paths, rather than genuine planning ability. Taken together, these failures demonstrate that current training regimes overemphasize correlation-driven visual-language alignments while underrepresenting embodied, counterfactual, and physics-based reasoning. Closing these gaps requires richer training distributions (e.g., embodied simulation data, physics-augmented learning) alongside architectural innovations that explicitly model reference frames and relational constraints. Absent such advances, models will continue to exhibit brittle competence that breaks down under compositional or physically grounded tasks.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models achieve the strongest overall performance in the open-ended evaluation, yet their results remain decidedly uneven across sub-categories. GPT-5-mini is the clear leader, consistently securing the best scores in **3D Geometry** and **Relative Positioning**, with peaks of 58.14% in Directional Relations and 55.81% in Alignment Patterns. Gemini-2.5-Flash and Claude 3.5 Haiku, by contrast, register very weak results in Proximity Gradients (16.28%) and Perspective Distortion (12.0%), respectively. This wide gap exposes the fragility of even the strongest models when reasoning over subtle, continuous visual cues. These disparities stem from the differential sophistication of models’ spatial reasoning pipelines, where some architectures capture relational and geometric structure far more effectively than others. Tasks such as Proximity Gradients are especially demanding, requiring analogical, graded reasoning rather than discrete classification. GPT-5-mini’s success where others fail establishes that scaling and proprietary training corpora confer distinct advantages. Even the leader, however, shows clear vulnerabilities, confirming a systemic issue: closed-source dominance is relative, not absolute, and these models still fall short of robust, human-like spatial reasoning.

Open-source Large Models (Table 13). Large open-source models produce a more distributed performance landscape, with different systems excelling in isolated sub-categories. llama-3.2-90b-vision-instruct achieves 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads in Spatial Containment at 38.0%. These highlights notwithstanding, several models collapse on Proximity Gradients, with qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct both scoring only 9.3%. This proves that architectural scaling helps but does not guarantee coverage across difficult continuous-reasoning tasks. Much of the variation traces to heterogeneous training data: models exposed to richer depth and geometry corpora dominate occlusion and containment tasks, while others generalize poorly. The persistent weakness on Proximity Gradients demonstrates that these systems are not optimized for subtle spatial analogies and distance scaling. Inter-model variance in this group is markedly higher than among closed-source models, underscoring the impact of open-source development heterogeneity. Large open-source models thus deliver encouraging peaks with inconsistent coverage, establishing that scale alone is insufficient without targeted pre-training and fine-tuning.

Table 12: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Table 13: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

Open-source Small Models (Table 14). Small open-source models predictably lag behind their larger counterparts, yet their results reveal noteworthy localized strengths. qwen-2.5-vl-3b-instruct achieves 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads in Spatial Containment (24.0%) and Stability Prediction (21.43%). These successes are decisively offset by catastrophic failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B manages only 2.0% in Shape Projection. This extreme variability is driven by limited capacity and restricted training diversity, which prevent small models from learning robust representations of geometric and relational principles. Sub-categories demanding precise reasoning about orientation and hierarchy trigger complete breakdowns, as small misinterpretations translate into total failure. That these models occasionally outperform larger ones in narrow tasks demonstrates that even small-scale training can instill isolated competencies. This confirms both the value of specialization and the structural ceiling of compact architectures. In short, small open-source models

Table 14: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

show pockets of promise but remain highly brittle, with overall accuracy bounded by size and data constraints.

Reasoning Models (Table 15). Reasoning-oriented models present a sharp internal divide, with Gemini-2.5-flash-thinking standing far above its peers. It records the best scores in nearly every sub-category, including Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14%. Kimi-VL-A3B-Thinking-2506, by contrast, struggles dramatically, scoring just 6.0% in Spatial Containment and trailing across the board. This gap arises from differences in how these models integrate multi-modal streams into coherent reasoning frameworks. Gemini-2.5-flash-thinking forms and queries internal world models, enabling it to infer object interactions and physical properties far more effectively, whereas weaker models rely on surface-level pattern matching that proves inadequate for compositional or causal reasoning. The consistently high performance of Gemini-2.5 across categories establishes that explicit reasoning mechanisms deliver tangible benefits for spatial understanding. Its failure to dominate Shape Projection and Betweenness Relationships nonetheless proves that even advanced reasoning modules are not yet universally effective. Reasoning models therefore hold clear promise, but the category itself remains heterogeneous and far from solved.

Spatial Reasoning Models (Table 16). Spatial reasoning models, though explicitly designed for the task, record the lowest results overall. Performance is severely fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct edges ahead with 23.91% in Volume Comparison and 13.73% in Scale Consistency, but these scores stay far below competitive thresholds. At the other extreme, Betweenness Relationships and Corner/Angle Positioning collapse entirely, recording 0.0% and 2.78%, respectively. These results prove that the models’ smaller capacity and narrowly focused training severely constrain their representational flexibility. The failures establish that narrow specialization without sufficient capacity or data breadth cannot yield generalizable spatial reasoning: on geometric and positional tasks in particular, even slight representational errors cascade into total breakdowns. Despite their focus, spatial reasoning models are therefore the least effective, underscoring that both scale and diversity are necessary for robust performance.

Overall Discussion. Across all categories, closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16), the pattern of fragmented competence and systemic weakness is consistent. Closed-source models perform best overall, with GPT-5-mini reaching highs of 58.14% in Directional Relations and 55.81% in Alignment Patterns, yet even they collapse on Proximity Gradients. Large open-source models post impressive peaks such as 60.0% in Depth and Occlusion but still fail at just 9.3% in Proximity Gradients, proving

Table 15: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Table 16: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

that scale without specialization remains insufficient. Small open-source models confirm this, pairing rare localized strengths (e.g., 26.0% in Relative Size Comparison) with catastrophic lows like 0.0% in Betweenness Relationships. Reasoning models establish that explicit reasoning mechanisms markedly improve results, with Gemini-2.5-flash-thinking reaching 75.0% in Tool Handedness, yet peers like

Table 17: Distribution of question counts by the number of models (0–5) that correctly answered each item across categories and sub-categories in SPATIALAB-OPEN. “0” indicates no model answered correctly, while “5” indicates only five models succeeded to answer these among all models tested.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

Kimi-VL-A3B fall to single digits, exposing the fragility of less-integrated reasoning frameworks. Spatial reasoning models, despite their specialization, underperform the most, with scores often below 15% and total failures on Betweenness Relationships. Taken together, these results isolate three root causes: (1) a lack of robust representations for continuous and analogical spatial cues, (2) structural limitations tied to scale and training diversity, and (3) brittle reasoning pipelines that prevent generalization across tasks. No current approach, whether proprietary scaling, open-source expansion, reasoning integration, or specialization, delivers comprehensive coverage. The frequent within-model gaps exceeding 50 percentage points further prove that progress is piecemeal. Future work must therefore unify robust geometric encodings, physics-informed reasoning, and cross-task transfer, moving beyond isolated gains toward consistent spatial intelligence.

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 exposes persistent weaknesses across categories, with a pronounced skew toward the lower bins (0–2 models correct). 3D Geometry is broadly challenging, with Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) showing especially poor outcomes that prove models lack reliable physical and geometric reasoning. Depth and Occlusion is the hardest category overall, accumulating the highest zero-correct counts (72), concentrated in Layering Order (20 zero) and Transparency Effects (16 zero), both of which demand nuanced handling of occlusion and non-local cues. Orientation struggles similarly: Cardinal Direction achieves some mid-range success, but Stacking Orientation (11 zero) and Tool Handedness (5 zero, none with perfect success) confirm the brittleness of embodied directional reasoning. Relative Positioning reveals pervasive inconsistency, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models, while Corner/Angle Positioning shows low but somewhat balanced performance. Size and Scale is comparatively stronger, with multiple subcategories (e.g., Scale Consistency with 6 full successes) demonstrating that some models internalize coarse perspective and scaling cues, though failures stay high in Shadow-Size Projection (17 zero). Spatial Navigation is split: Spatial Sequence and Obstacle Avoidance yield occasional full-model success, but Pathway Existence (16 zero) and Accessibility Constraints (10 zero) expose persistent difficulty in multi-step relational chaining. The grand total makes the verdict decisive: across 356 items, zero-correct dominates (356), while only 74 items are solved by all five models, confirming that consistent spatial competence remains elusive.

These findings establish that the root causes of failure stem less from dataset design and more from the representational and reasoning limitations of current models. First, the dominance of errors in Depth and Occlusion reflects a reliance on superficial appearance features rather than structured scene representations: models latch onto surface textures instead of reasoning about hidden layers, producing systematic errors in Layering Order. Failures in Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, since pretrained internet imagery rarely carries explicit annotations of left/right handedness or stacking physics. The brittleness in Relative Positioning and Spatial Navigation reveals clear deficiencies in chaining multi-hop spatial relations; Pathway Existence, for instance, requires integrating local connectivity with global layout, which attention-based co-occurrence priors fail to capture. Partial success in Size and Scale, by contrast, demonstrates that models exploit statistical regularities in perspective distortion or scale consistency, only to collapse under less frequent cases like shadows. Collectively, these patterns prove that current models succeed when cues are salient and correlate with training priors but fail whenever reasoning demands counterfactual, compositional, or embodied inference. Closing these gaps requires architectural advances that explicitly model reference frames, occlusion layers, and physical dynamics, together with training data enriched by embodied simulations and counterfactual augmentations. Absent these, progress in spatial reasoning will remain fragmented, with models performing well on surface-level cues but breaking down on structured, real-world spatial logic.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

In this section we present a detailed analysis of the performance gap between multiple-choice (MCQ) and open-ended evaluation formats across spatial reasoning categories in SPATIALAB, as demonstrated in Table 18.

G.1 QUANTITATIVE SNAPSHOT

Across 25 models, the average performance gap between MCQ and open-ended formats is **23.0%** with a standard deviation of 5.5%. Dataset-wide means per subtask are: 3D geometry (24.57), depth & occlusion (23.45), orientation (23.70), relative positioning (23.11), size & scale (23.42), and spatial navigation (22.89), yielding an overall average of 23.01. The overall gap spans a minimum of 11.93 to a maximum of 31.57 (median $\approx$ 23.34). At the family level, spatial reasoning models exhibit the largest average gap (27.03), while reasoning-oriented models show the smallest (19.11). Proprietary models carry lower gaps on average (21.09) than many long open-source models (24.51), and the human baseline records an overall gap of 22.64. The highest gaps are InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%); the lowest are Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlational analysis

Table 18: **Performance gap between MCQ and Open-ended (%) ($\downarrow$) across spatial categories on SPATIALAB.** A lower value means the model performs comparably in both setups, whereas a higher value marks a sharp disparity. We bold and underline the largest gap within each model group.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

establishes that spatial navigation dominates the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also strongly aligned.

G.2 PATTERNS AND OBSERVATIONS

Three broad patterns emerge. First, specialist spatial reasoning models exhibit the largest MCQ→open-ended gaps, establishing that they are optimized for classification-style supervision and less robust to free-form generation. Reasoning-focused models, by contrast, display smaller gaps, evidencing the stabilizing effect of chain-of-thought–style training on generative outputs. Second, spatial navigation is the strongest predictor of overall disparity: models that falter on open-ended navigation reasoning drive the aggregate gap, with orientation and 3D geometry subtasks contributing substantially and relative position and size/scale varying more idiosyncratically across models. Third, scale alone does not explain robustness: smaller models such as Llama-3.2-11B achieve among the lowest gaps, while very large models such as InternVL3.5-72B show the highest. Finally, several negative or near-zero gaps appear (e.g., depth & occlusion for Llama-3.2-11B at -1.98 , relative position for o4-mini at -3.18), proving that open-ended evaluation is in some cases more reliable than MCQ, driven by misleading distractors.

G.3 HYPOTHESES ON ROOT CAUSES

We identify several interacting factors that underlie the observed MCQ→open-ended performance gaps in SPATIALAB, each supported by statistical observations:

1. **MCQ structural advantage.** The MCQ format constrains the output space, letting models exploit surface cues or eliminate distractors. Across models, the mean MCQ score is systematically higher than the open-ended score, with the average gap at 23.0% and $\sigma = 5.5\%$. The negative gaps for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) prove that open-ended generation can outperform MCQ when distractors are misleading, confirming that MCQ artificially inflates or distorts measured performance.

Recommendation: We recommend complementing MCQ evaluations with open-ended tasks and subtask-level diagnostics, which capture model competence more accurately and eliminate the overestimation caused by option-based cues.

2. **Specialization bias in spatial reasoning models.** Spatial reasoning models exhibit the largest average gaps (27.03%) despite being optimized for spatial tasks, establishing that they were trained primarily on categorical or synthetic selection tasks rather than free-form generation. InternVL3.5-72B, a 72B parameter spatial reasoning model, posts the highest overall gap (31.57%), concentrated in spatial navigation (36.68%) and orientation (34.44%), which exposes a clear reliance on classification-like supervision. Correlations between subtask gaps and overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) confirm that these format-dependent failures cluster in sequential and geometric reasoning tasks.

Recommendation: To eliminate this bias, we recommend fine-tuning or instruction-tuning on open-ended spatial generation tasks, which improve robustness to generative formats and cut reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models (e.g., Gemini-2-Flash-Thinking, o4-mini) demonstrate smaller average gaps (around 19.11%) and lower variance across subtasks. Gemini-2-Flash-Thinking, for instance, records an overall gap of 13.0% and consistently modest subtask gaps (Difference in 3D Geometry: 13.9%, Difference Relative Positioning: 21.03%), proving that chain-of-thought or stepwise decoding drives stable open-ended responses. The reduced correlation between subtask gap variance and overall gap within this family confirms that instruction-tuned reasoning suppresses format sensitivity.

Recommendation: To counter format sensitivity, we recommend adopting chain-of-thought or stepwise decoding strategies and instruction-tuning for open-ended spatial tasks, which improve generative consistency and reduce MCQ dependence.

4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps are the strongest predictor of overall MCQ→open-ended disparity (Pearson $r = 0.99$). Models with high spatial-navigation gaps, such as SpaceQwen2.5-VL-3B-Instruct (Spatial Navigation gap = 35.87%), also carry large overall gaps (29.78%), establishing that multi-step referential reasoning is disproportionately affected in open-ended formats. MCQ options thus simplify the reasoning chain, masking underlying weaknesses in sequential grounding and relational computation.

Recommendation: To capture sequential reasoning abilities faithfully, we recommend including dedicated open-ended spatio-navigation tasks and targeted multi-step reasoning supervision, which strengthen grounding and remove the overestimation introduced by MCQ.

5. **MCQ distractor artifacts.** Negative and near-zero gaps prove that poorly designed distractors suppress measured MCQ performance. o4-mini, for instance, shows a negative relative position gap (−3.18%) while maintaining positive gaps in other subtasks, demonstrating that distractor quality artificially depresses MCQ accuracy. Conversely, models like Qwen-VL2.5-32B-Instruct exhibit very high MCQ→open gaps (overall = 29.85%) across multiple subtasks, establishing that MCQ distractors also exaggerate perceived competence in selection-based tasks.

Recommendation: We recommend careful auditing and refinement of MCQ distractors alongside open-ended evaluation, which together yield accurate, unbiased measures of model spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

Our findings establish the necessity of reporting both MCQ and open-ended results. Exclusive reliance on MCQ inflates estimates of practical spatial competence, with average disparities around 23%. Subtask-level breakdowns are essential, since spatial navigation disproportionately drives the observed differences. Diagnostic evaluations that disentangle classification from generative reasoning isolate error modes far more precisely. Instruction-tuning and decoding strategies that encourage stepwise reasoning are a proven lever for reducing MCQ dependence. Finally, careful design and auditing of MCQ distractors are indispensable for fair cross-format comparisons.

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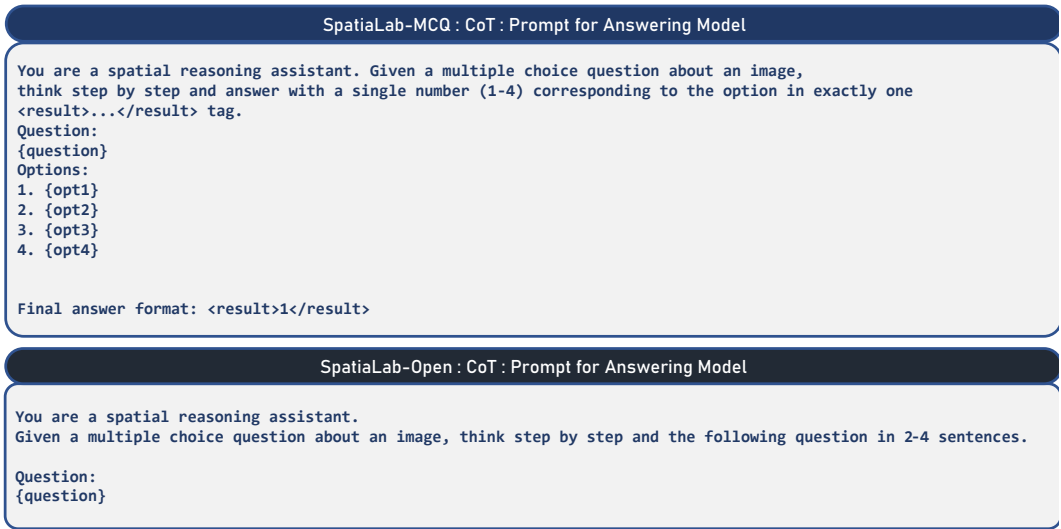


Figure 7: Chain-of-thought (CoT) Prompts.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To enhance spatial reasoning performance on our diverse visual question benchmark, we systematically evaluated multiple complementary strategies and pinpoint exactly where each one succeeds and where it fails.

H.1 INHERENT REASONING MECHANISMS

We next quantify the role of built-in reasoning mechanisms by comparing standard models with their reasoning-enabled counterparts (Table 2, Table 3). Across MCQ evaluations, reasoning consistently boosts performance, with the magnitude scaling by category. For instance, Gemini-2.5-Flash-Thinking improves over its non-reasoning variant by +4.64% overall (52.93% vs. 48.29%), driven by decisive gains in Relative Positioning (+13.1%) and Orientation (+8.6%). The same pattern holds for o4-mini-medium, which reaches the highest overall MCQ accuracy (53.21%), establishing that explicit reasoning scaffolds are especially effective in structured, discrete-answer settings.

Open-ended evaluation reveals a sharper, more selective picture. Here, reasoning-equipped models again outperform their baselines, but the gains are unevenly distributed. Gemini-2.5-Flash-Thinking edges past Gemini-2.5-Flash (32.77% vs. 30.93%), while the decisive leap belongs to o4-mini-medium (37.86%), which significantly surpasses every non-reasoning model. These results establish that reasoning mechanisms mitigate the instability of free-form generation, especially in complex categories such as 3D Geometry and Spatial Navigation. Crucially, the gains are not universal: reasoning models still regress in specific categories, e.g., Gemini-2.5-Flash-Thinking drops sharply in Size & Scale (21.74% vs. 26.59%), confirming our earlier finding that generative settings amplify brittleness.

Overall, these comparisons establish that explicit reasoning modules substantially enhance structured MCQ performance and stabilize open-ended responses, while their benefits remain uneven across categories. Reasoning mechanisms demonstrably strengthen logical consistency and relational chaining, yet they do not resolve the core challenges of scale sensitivity, occlusion, and free-form linguistic grounding.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To determine whether structured reasoning improves spatial performance, we evaluated models under Chain-of-Thought (CoT) prompting. For each of the 30 sub-categories in SPATIALAB, we sampled 5 items (seed fixed to 42) to construct a balanced test set covering all major spatial phenomena. This design guarantees fair coverage while keeping evaluation computationally tractable. Example CoT

prompts are shown in Figure 7, which illustrate the step-by-step reasoning scaffolds provided to the models. The setup cleanly isolates the contribution of CoT reasoning from model scale or training.

Table 19: Performance with CoT prompting across categories for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%) and Gains represent the difference between with and without CoT prompting.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

Our analysis (Table 19) demonstrates that orientation is the only category where CoT prompting reliably improves performance, while every other category either stagnates or declines. This proves that reasoning helps only when the task reduces to logical alignment of directions, and fails outright when richer perceptual priors (e.g., depth, scale, relative positioning) are required. The cause is clear: models lack inherent knowledge of 3D perception and geospatial relationships and are not fundamentally “aware” of these concepts in the way humans are. Consequently, CoT prompting does not correct misconceptions but reinforces them, propagating and amplifying errors across reasoning steps.

This finding establishes that our benchmark is non-trivial: it cannot be solved by bolting multi-step reasoning onto existing models. Unlike textual reasoning tasks where CoT reliably improves outcomes, our benchmark exposes deep architectural limitations in perceptual and spatial grounding. In categories like depth and size, incorrect priors guarantee that every additional reasoning step moves the model further from the correct solution. The sharp performance drops, particularly for Gemini in MCQ settings, prove how fragile reasoning becomes when it is not anchored by robust perceptual knowledge.

Thus, our task design is a decisive strength: it goes beyond surface-level reasoning and directly probes whether models possess genuine understanding of geospatial and 3D relational concepts. Solving it demands more than chaining logic; it requires embeddings and training that capture core aspects of perception and spatial cognition. Our benchmark therefore challenges models at a fundamental level and exposes a critical gap in current architectures.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We extend the CoT setup with a self-reflection stage, where the model reviews and revises its initial reasoning before finalizing an answer (see Figure 8). The evaluation protocol remains identical to the standard CoT setting, using 5 balanced samples per sub-category (seed 42). This design cleanly isolates the effect of self-reflection while keeping the comparison directly aligned with baseline CoT performance.

Table 20: Performance with CoT vs. CoT + Self-Reflection (CoT-SR) across categories for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%) and Gains represent the difference between CoT-SR and CoT.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

Adding self-reflection to Chain-of-Thought (CoT) prompting produces sharply model-dependent outcomes. For InternVL-3-78b, overall accuracy declines (-2.66% MCQ, -8.67% open-ended),

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Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

with sharp drops in depth, relative positioning, and navigation, demonstrating that reflection reinforces errors rather than correcting them. A few localized gains appear, such as +12% in Size & Scale (MCQ), but they are inconsistent and overwhelmed by large degradations (−28% in relative positioning open-ended). In contrast, Gemini-2.5-Flash delivers strong MCQ gains (+20.66%), with decisive improvements in 3D Geometry (+32%) and Depth & Occlusion (+24%), while open-ended improvements remain negligible (+0.67%). These results expose a key asymmetry: reflection helps in multiple-choice contexts where models can prune unlikely options, but does not transfer to free-form generation, where uncertainty propagates into ungrounded rationales. The categories most improved by reflection are precisely those where geometric consistency is central (orientation, depth, size), whereas tasks requiring relational integration across multiple objects (relative positioning, navigation) degrade. In sum, reflection acts as a stabilizer in structured-choice settings but fails to enhance naturalistic reasoning.

This inconsistency mirrors our findings with plain CoT prompting: reasoning enhancements succeed only when the model has stable perceptual anchors to ground its steps. Self-reflection adds an explicit verification pass, but this mechanism is only as strong as the model’s internal priors. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies misconceptions rather than repairing them. This directly explains why Gemini, with stronger perceptual grounding, benefits under MCQ conditions, while InternVL-3-78b degrades more frequently. The root cause is a pretraining mismatch: current VLMs import reasoning templates from language modeling but lack the embedded 3D and geospatial representations that humans leverage. Without these anchors, reflection cannot perform true error correction; it operates as a linguistic filter that polishes reasoning syntax without improving perceptual grounding. Reflective reasoning loops

therefore provide modest utility in discrete answer formats but collapse under open-ended demands where factual verification is absent.

These findings confirm that our benchmark exposes a fundamental gap in current architectures: surface-level reasoning layers cannot compensate for missing spatial cognition, and meaningful progress requires deeper integration of geometric and perceptual structure. Moreover, because VLLMs are trained predominantly on web data, the concepts well-represented online are exactly those where models perform better, whereas poorly represented spatial and 3D concepts suffer, which directly reinforces these performance disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) using a parameter-efficient approach with LoRA (Hu et al., 2021). To operate within limited GPU resources, we loaded the model in 4-bit precision (load_in_4bit=True) and inserted LoRA adapters into attention and feed-forward network projection layers. The LoRA hyperparameters, including rank, scaling factor, dropout, and target modules, are summarized in Table 21. Gradient checkpointing reduced memory usage, and a fixed random seed guaranteed reproducibility.

For tokenization, we followed the Qwen-2.5 chat template to enforce consistent formatting between training and inference. Each training example was converted into a single text sequence, and dynamic padding was applied with DataCollatorForSeq2Seq. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4). Trainer and optimization hyperparameters, including learning rate, optimizer, weight decay, and scheduler, are listed in Table 22. The training loop ran four passes, each spanning a single epoch, yielding four saved checkpoints.

This setup efficiently fine-tunes a large vision-language model on limited GPU resources while retaining high adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a reveal a sharp divergence between the MCQ and Open-ended evaluation modes over training epochs. MCQ accuracy begins at a lower baseline of 25.63% but rises consistently, converging near 36.59% by the fourth epoch. This steady improvement demonstrates that the model benefits strongly from additional supervised fine-tuning, particularly in tasks that constrain answers within a limited candidate space. By contrast, Open-ended evaluation starts higher at 34.40% but collapses by the second epoch, dropping to as low as 12.62%. Only in later epochs does it partially recover, reaching 35.48% at the end. This instability exposes a sensitivity in open-ended generation that is absent in the MCQ setting, where the structure of multiple-choice questions supplies more stable optimization signals. We attribute the fluctuations in open-ended performance to fine-tuning overfitting to MCQ-like reasoning patterns at the expense of free-form language generalization. The dotted baselines in the plot make clear that the MCQ mode achieves gains well beyond its initial standing, whereas Open-ended remains essentially stagnant. The figure establishes that SFT decisively favors MCQ tasks in both stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure with a detailed before-and-after breakdown across six spatial reasoning categories. Across MCQ tasks, performance improves in every category, with gains ranging from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale. The Grand Total gain for MCQ reaches 10.97%, proving that the fine-tuning procedure effectively enhances structured-answer performance. In contrast, Open-ended results are markedly mixed, with both positive and negative changes. Categories such as 3D Geometry and Spatial Navigation show small positive gains (3.50% and 2.82%, respectively), yet Size and Scale drops sharply by -3.95% and Relative Positioning decreases by -0.79% , showing that fine-tuning actively harms certain reasoning modes in free-form contexts. The overall Open-ended gain is only 1.07%, negligible against the MCQ improvements. These results establish that the effectiveness of SFT is decisively governed by the evaluation paradigm. By isolating the categories, we pinpoint exactly where training helps, where it fails, and how benefits diverge between constrained and unconstrained outputs.

A deeper inspection reveals striking disparities between categories and formats. For MCQ, Size and Scale and Spatial Navigation yield the highest improvements, demonstrating that models trained with multiple-choice prompts learn scale discrimination and navigational reasoning especially well. However, in Open-ended form, Size and Scale declines, proving that the same learned representations fail to transfer to generative contexts. Relative Positioning barely improves in MCQ and worsens in Open-ended, exposing a consistent weakness in relational reasoning regardless of format. Conversely, Orientation shows strong MCQ gains (+13.33%) and small Open-ended improvements, establishing that some categories generalize more robustly. The divergence between the two formats stems from the structural scaffolding MCQs provide, which removes the need for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, demand simultaneous mastery of reasoning and natural language fluency, which amplifies weaknesses. The trends make clear that while fine-tuning strongly enhances pattern recognition in discrete-choice settings, it introduces volatility and outright degradation when free expression is required. This duality demonstrates that training interventions optimized for MCQs do not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we establish that SFT as currently applied delivers strong benefits for structured tasks but limited or outright negative transfer to open-ended reasoning. This limitation is not merely an artifact of fine-tuning; it is symptomatic of how contemporary VLMs are pretrained. Existing pretraining pipelines heavily emphasize broad image-text alignment, object recognition, and captioning, yet provide little explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. Consequently, models excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning demonstrates that the model’s learned priors are fragile: without multiple-choice scaffolding, they revert to shallow pattern-matching instead of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, expose missing inductive biases in pretraining, biases that humans develop through embodied interaction with 3D environments. A further contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, preventing models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to rethink both pretraining and fine-tuning for spatial reasoning, incorporating

datasets and objectives that reflect naturalistic visual-spatial challenges. Ultimately, without targeted reforms, current VLMs remain locked in an algorithmic straightjacket: they appear competent under constrained evaluations yet stay brittle and unaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To establish the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a decisive divergence in performance dynamics between Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As shown in Table 23 and Figure 9, performance on SPATIALAB-MCQ improves monotonically, rising from a baseline of 25.63% to an average of 35.74% at Epoch 4. This proves that the model successfully adapts to the discriminative nature of the task, mapping visual features to constrained output options with increasing reliability.

In sharp contrast, SPATIALAB-OPEN displays a pronounced *U-shaped* trajectory across all seeds. Instead of steady improvement, the model suffers an immediate and severe performance collapse in the early epochs (dropping from 34.40% to an average of 19.26% at Epoch 1). Although performance recovers in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible (+0.42%).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). MCQ performance improves steadily, whereas Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, exposing representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

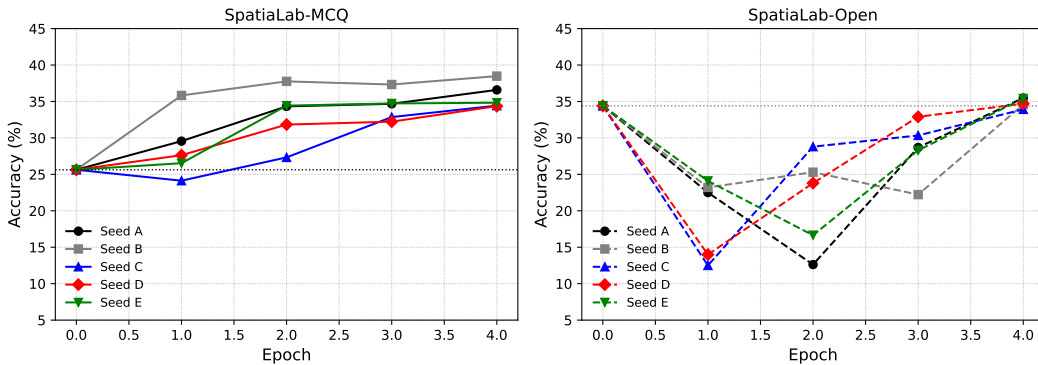


Figure 9: SFT Dynamics Analysis.

Representational Brittleness and Alignment Tax. This phenomenon is driven by the fragility of spatial representations in current VLMs under fine-tuning. The initial collapse in open-ended accuracy is *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation. As the model optimizes for the specific instruction-following format of the training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible, generative spatial grounding required for open-ended descriptions.

The subsequent recovery phase is the model re-learning to articulate spatial concepts within the new distribution, yet it fails to surpass the initial baseline. This dynamic establishes a critical finding: SFT is effective for teaching models to *select* correct spatial answers, but it does not enhance the underlying *generative spatial representation* needed for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To determine whether the representations learned from SPATIALAB generalize beyond the benchmark’s specific format, we evaluated the transferability of our fine-tuned model to three external spatial reasoning datasets: OMNISpatial, SPACE (Multimodal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains prove that the model learns generalizable spatial representations rather than overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As shown in Table 24, fine-tuning on SPATIALAB delivers substantial and consistent performance improvements across all evaluated domains, ranging from +5.24% to +11.70%. The largest gain lands on MIND THE GAP (+11.70%), demonstrating that our benchmark’s emphasis on occlusion and spatial continuity directly addresses the reasoning gaps targeted by that dataset. The improvements on OMNISpatial (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded in our taxonomy align with broader definitions of spatial intelligence in the literature. These results validate the quality of the SPATIALAB data. While the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the strong transferability proves that the dataset does not encourage overfitting to a specific question template. Instead, the diversity of the 30 sub-categories forces the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIOLAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each managed by a dedicated agent, enabling focused processing of objects, attributes, relations, symmetries, and transformations. This modular design targets accuracy and interpretability while supporting multi-step reasoning, propagating low-level perceptual cues through higher-order relational and transformational inferences. The architecture, agent implementation, and their coordinated interaction are detailed in the following sections, with the full set of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. The Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the

prompt enforces exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach retains even subtle visual features, such as partial occlusions or minor shape variations, for later relational and transformational reasoning.

Object Segmentation Agent. The Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent also surfaces overlaps, repetitions, and hierarchical groupings that are not evident in raw descriptions.

Attribute Extraction Agent. The Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults guaranteeing robustness against model inconsistencies. This structured encoding enables quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content while normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to eliminate relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure supports reasoning over both pairwise interactions and complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. The Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to preserve downstream compatibility. This agent captures emergent scene regularities that are invisible at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding lets the system track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent links transformations to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing `objects`, `relations`, `groups`, `symmetries`, and `transformations`. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to guarantee pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent produces a

comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs comprise objects, attributes, relations, groups, symmetries, and transformations. The model integrates geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For our technical implementation, we use Gemini-2.5-Flash with a low temperature (0.1) to enforce deterministic outputs and perform multiple iterations for refinement. API calls are handled via OpenRouter, supporting both text and image inputs, while parsing of model responses relies on regex and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition focuses each specialized model on a constrained sub-task, targeting accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to the standard CoT and CoT-SR setting, using 5 balanced samples per sub-category (seed 42), for a total of 150 samples.

Table 25: Agent Performance Analysis

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

The evaluation of Gemini-2.5-Flash with agents exposes both the strengths and the hard limits of agentic reasoning in perceptual-geometric tasks. Orientation is the only category where agents deliver consistent and substantial gains, improving by +8.00% in MCQ and a decisive +36.00% in open evaluation. This proves that agent-based multi-step reasoning is effective when the task decomposes into sequential alignment of directional cues, where structured deliberation clearly outperforms single-step outputs.

In contrast, most other categories exhibit negative or neutral gains, proving that agent reasoning does not universally enhance performance. Depth & occlusion suffers a severe -24.00% drop in open evaluation, while spatial navigation decreases by -12.00%. These tasks demand deep perceptual grounding in three-dimensional space, and the absence of such priors means reasoning steps reinforce misconceptions rather than correct them. Relative positioning likewise shows stagnation in MCQ and a -14.00% decline in open evaluation, demonstrating that agent loops add no robustness when relational understanding is inherently weak.

A few categories show marginal improvements, such as size & scale (+4.00% in MCQ), but these gains are unstable and do not transfer to open evaluation. Even in 3D geometry, agents provide no boost, with -4.00% and -12.00% drops in MCQ and open evaluation, respectively. These results establish that our benchmark is complex and cannot be solved by layering multi-step agentic reasoning

on top of existing models. It requires deeper integration of geospatial understanding and perceptual representations, which current architectures lack.

Overall, these results prove that agentic workflows alone are insufficient for advancing performance in this domain. While orientation tasks benefit from structured reasoning, most categories expose the models’ lack of inherent awareness of perceptual concepts. Errors then propagate through multi-step reasoning, amplifying performance drops. Our benchmark therefore serves as a rigorous stress test, establishing that future progress demands embedding genuine perceptual and geospatial priors rather than relying on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section presents a focused qualitative analysis of model failures observed in our visual reasoning benchmark. We explain each error class, deliver diagnostic root-cause analysis, identify recurring patterns, and describe verification protocols. The discussion is both interpretable and actionable, directly informing future architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models achieve strong coarse perception: they reliably recognize object categories, colors, and simple binary relations, but they break down systematically the moment multiple cues must be integrated. These failures are not random noise; they cluster into a small set of interpretable classes: spatial mislocalization, perspective and scale mistakes, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, proving that the underlying causes are common inductive biases in current vision–language pipelines rather than isolated training artifacts. Critically, the same error appears in both large-scale proprietary systems and small open-source baselines, exposing structural weaknesses shared across the entire model family. Below, we unpack each error class with representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** When queries require distinguishing one object from a set of visually similar candidates, models frequently misidentify the target. For example, when asked to point out “the leftmost red cube among three stacked cubes,” the model outputs the middle cube, driven by lighting highlights or central positioning. This problem is most acute in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. The model’s errors are not random: it privileges objects that appear most salient due to size, brightness, or centrality, even when these cues directly contradict the explicit spatial instructions in the query. These consistent patterns establish that the failures are systematic, not accidental.

★ **Analysis.** The recurring mistakes demonstrate that models lack a robust object-centric representation, or what cognitive science calls an “index” that persists across reasoning steps. Rather than maintaining a stable pointer to the correct candidate throughout the reasoning chain, models drift, reassigning attention to whichever object appears most visually striking at intermediate layers. This produces reasoning errors such as starting with the leftmost cube but ending the chain with the brightest or largest cube. The inability to anchor spatial reference consistently makes tasks involving relative comparisons, like “the ball behind the second chair,” especially fragile. In essence, the model operates as a salience detector, not a referential reasoner.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether via attention or convolution, mix contextual information from neighboring regions. While beneficial for global context, this erases the fine-grained, object-specific identity needed to distinguish similar candidates. Once pooled, the features of two adjacent red cubes become nearly indistinguishable in latent space. Without an explicit mechanism such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation prevents reliable referent tracking across reasoning steps and directly explains why models consistently confuse dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks involving metric reasoning about relative size, fit, or scale are handled inconsistently. On simple comparisons like “is a car larger than a bicycle,” models succeed because such relationships align with strong statistical priors. But on near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” errors become frequent and unpredictable. A recurring pattern is reliance on canonical object sizes: predicting that a door must be taller than a truck even in a rendering where the truck is scaled down. Failures also occur in unusual visualizations, such as toy miniatures or architectural models, where familiar proportions are deliberately inverted. In these cases, models cling to their priors and ignore the actual geometric evidence visible in the image.

★ **Analysis.** This behavior proves that models rely on distributional statistics of object categories rather than performing true geometric inference. Instead of reasoning about projection geometry, relative bounding box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut works in everyday contexts but fails catastrophically in edge cases that deviate from the training distribution. When an unusually small airplane is rendered next to a person, the model still predicts the airplane as larger. Such errors expose the lack of grounding in pixel-level metric cues and the absence of any consistent use of perspective or vanishing points.

⊗ **Root cause.** Current architectures include no explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object category labels and size but never the underlying geometry. Moreover, training objectives prioritize overall accuracy over fine-grained metric consistency, leaving scale-sensitive errors under-penalized. Without inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors about canonical sizes instead of interpreting the actual perspective geometry in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front, behind, or partially hidden remains a core weakness. Models misreport the foreground object when two animals overlap, or when semi-transparent barriers like glass or fences are involved. In an image where a cat sits partially behind a curtain, models incorrectly describe the curtain as occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are the most error-prone: they are frequently omitted entirely, producing wrong answers to questions like “what object lies behind the fence.” These mistakes occur consistently, especially when occlusion cues are subtle or the depth difference is small.

★ **Analysis.** These failures prove that models treat depth and occlusion as weak, secondary signals. Although the pixel data contains clear occlusion cues (e.g., edge boundaries, T-junctions), the intermediate features learned by the model do not retain them at sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, erasing the very cues necessary to resolve ordering. This is why models succeed at gross depth distinctions (foreground vs. background) but collapse in fine-grained cases where partial occlusion must be inferred. In practice they guess rather than infer, with a systematic bias toward treating the visually larger or brighter object as foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and provides no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are lost when features are pooled across large receptive fields. Furthermore, training data underrepresents subtle occlusion cases, especially those involving transparency or fine boundaries. As a result, the learned representations flatten layered scenes into 2D appearance maps, preventing robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A classic example is misjudging whether a crosswalk is safe: models base the answer solely on the presence of painted stripes while ignoring the state of the traffic light or approaching vehicles. Likewise, a surface with a metallic texture is misclassified as “steel” even when it is painted plastic. In tool-use contexts, the presence of handles or familiar shapes leads the model to infer functionality the object does not possess. These confusions are frequent in tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors prove that language priors dominate the model’s predictions when visual

input is ambiguous. The decoder favors plausible narratives grounded in training-data co-occurrence: “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so on. This shortcut lets models produce semantically coherent answers without inspecting the relevant visual evidence. The result is superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios, manifesting as overfitting to textual stereotypes rather than grounding in perceptual detail.

⊗ **Root cause.** The underlying training objective, maximizing the likelihood of fluent and plausible text, does not enforce visual grounding. As long as the generated answer is semantically reasonable, the loss is minimized even when the reasoning is wrong. The model is therefore incentivized to substitute common-sense priors for genuine perceptual inference. Without auxiliary losses or verification steps that tie textual claims to image regions, models will continue to swap perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models generate coherent but unsupported narratives. When asked “why is the object leaning,” a model answers “because the surface is slippery” even though slipperiness cannot be visually inferred. A tilted shelf is explained as “due to an earthquake,” a claim entirely outside the evidence in the image. These errors expose a consistent tendency to fill gaps with imaginative but ungrounded rationales, producing fluent explanations that mislead users about the true evidence.

★ **Analysis.** The issue stems from a decoupling between text generation and visual grounding. The model generates high-quality language that reads convincingly, but no mechanism ties these rationales to actual visual features. Confidence calibration compounds the problem: models assign high certainty to fabricated explanations, creating a false impression of reliability. As a result, open-ended outputs look authoritative while being factually baseless, a failure mode that is especially dangerous in safety-critical domains where plausible but false rationales misguide decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence but not factual grounding. Decoders are optimized to produce the most probable text continuation given the prompt, with no constraint linking the explanation back to specific image evidence. The absence of an explicit verification layer or grounding requirement forces models to default to linguistic plausibility over perceptual truth. Without additional architectural or training constraints, open-ended rationalizations will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

Comparing failures across different models exposes several consistent patterns that reveal the underlying mechanisms of errors. Disagreements between models are systematic, not random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relationships, while others fall back on linguistic priors or common-sense assumptions derived from training text. This produces predictable divergences, for example when two models give different answers about which object is “in front” despite seeing the same scene. Errors also spike when multiple cues must be integrated simultaneously. Tasks requiring reasoning about occlusion, relative scale, and perspective together are the most challenging; determining whether a larger object partially occluding a smaller one is in front or behind consistently triggers mispredictions. Calibration is a further failure: high-confidence predictions frequently correspond to incorrect answers, especially in open-ended explanations or complex spatial queries. These patterns prove that errors arise from how models balance perceptual grounding against linguistic plausibility. Different architectures exhibit distinct tendencies, some over-relying on visual geometry, others overfitting to language correlations, so failures are not isolated mistakes but reflect deeper trade-offs in reasoning strategy. Consequently, any attempt to improve performance must address both sources of bias simultaneously rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

These error patterns point to several targeted strategies that reduce failures without requiring a complete architectural redesign. Geometry-aware supervision, added through auxiliary tasks like depth prediction, occlusion ordering, or overlap estimation, forces models to respect spatial constraints

and improves consistency across predictions. Multi-scale feature retention offers a second lever: edge-aware pooling or attention mechanisms preserve information from thin or partially occluded structures that standard downsampling discards. Coordinate anchoring resolves confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Counterfactual augmentation, in which distractor objects are systematically swapped or perturbed, forces the model to rely on true referent cues rather than heuristics. Lightweight verification modules layered on top of existing outputs cross-check claims against visual or geometric constraints to filter implausible predictions. Each of these strategies is implementable incrementally, providing clear paths to reduce systematic failures while keeping the core model largely unchanged.

Beyond these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway to instill more reliable spatial reasoning. By curating datasets that explicitly highlight failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference-resolution traps, SFT directly exposes models to the edge scenarios where they otherwise default to heuristics. Carefully designed instruction-style fine-tuning drives step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than skip to a guess. This strengthens grounding and improves interpretability, since errors become traceable to specific reasoning steps. SFT can further be paired with counterfactual or adversarial augmentations, ensuring the model learns invariances to distractors and context shifts rather than memorizing. Critically, these interventions require no wholesale architectural changes: they adapt existing models by reshaping their inductive biases through supervision. SFT therefore serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In this work, we additionally performed SFT using 40% of the training data and obtained clear performance gains, particularly on multiple-choice questions. This demonstrates that targeted supervision helps models internalize spatial cues and resolve distractor confusions when answer options are well-defined. Improvements on open-ended responses, however, remained limited: the model continues to produce fluent but weakly grounded explanations, proving that SFT alone cannot resolve rationalization without verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers to achieve balanced progress across both structured and free-form tasks.

Our analysis proves that failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than random noise. VLM errors follow systematic trajectories tied to the inductive biases of their architectures and training regimes. Geometric reasoning collapses when occlusion cues are weak, and language priors then take over, producing answers that sound plausible but lack visual grounding. Scale inference drifts when canonical object-size priors dominate over pixel-level metric cues, yielding consistent misjudgments in near-threshold scenarios. These findings establish that the gap is not one of capacity but of representation: current models lack explicit mechanisms to anchor visual features to grounded geometric relations.

Adopting the proposed verification protocols and targeted mitigation strategies addresses these weaknesses directly. Geometry-aware supervision and counterfactual augmentation make models resist shortcuts, while multi-scale feature retention ensures thin structures and partial occlusions are not discarded. SFT, carefully applied, strengthens structured reasoning by aligning model predictions with task constraints, though it must be combined with grounding objectives to avoid overfitting to discrete-choice formats. Verification modules provide a lightweight way to enforce consistency between claimed relations and image evidence, counteracting the tendency to rationalize without checking. Together, these steps chart a practical pathway toward systems that reason more like humans: robust to clutter, interpretable in their failures, and capable of explaining not only what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{"ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position": "Leftmost"}},
 {{"ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position": "Center"}}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]}

```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 sub-categories in SPATIALAB, illustrating the full breadth of spatial reasoning challenges covered. These samples span the variety of visual phenomena, including 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation tasks. Together, they give a concrete view of the reasoning each model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- **GPT-5-mini:** Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- **Gemini-2.5-Flash:** The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- **InternVL3.5-72B:** From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

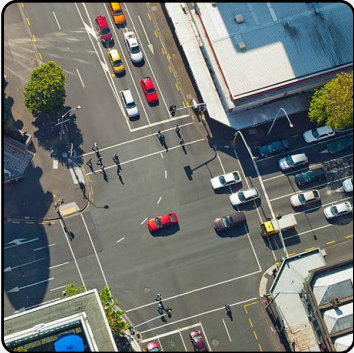
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position. The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

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Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

1. Up
2. Down
3. Away
4. Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- **GPT-5-mini:** If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- **Gemini-2.5-Flash:** The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- **InternVL3.5-72B:** If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- **Kimi-VL-A3B-Thinking-2506:** The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

1. 90
2. 45
3. 180
4. 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

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Relative Positioning : Proximity Gradients



Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- Gemini-2.5-Flash: "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- InternVL3.5-72B: The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- Kimi-VL-A3B-Thinking-2506: Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Some examples from the benchmark (Relative Positioning: Proximity Gradients) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

To establish that the benchmark results are not artifacts of random initialization or dataset noise, we evaluate both model robustness across runs and dataset stability across repeated trials. Such analyses are essential for small- to medium-scale benchmarks, where minor fluctuations can produce misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of the same model. Each run uses a distinct random seed while preserving all other hyperparameters.

K.1 MODEL ROBUSTNESS

To rigorously establish model evaluation robustness, we perform several statistical studies, as described below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify agreement between runs, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979), using a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ is the variance between items and σ_{within}^2 is the residual variance across runs. ICC values above 0.75 indicate good reliability, while values above 0.9 establish excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We also analyze the stability of the dataset itself, testing whether question difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further assess **robustness**, we perform a resampling analysis. From the total of $C = 30$ subcategories, we randomly sample subsets of size $S \in 20, 25$ for each sub-category from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, the Wilcoxon signed-rank test assesses median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ be the difference in accuracies for run r , excluding ties ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we fail to reject H_0 , confirming that the accuracies from 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty can be defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, the correlation between runs should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) establishes stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To measure overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ indicate excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the analysis on two models: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby spanning both large- and small-scale models as well as proprietary and open-source paradigms. We conduct all tests in both multiple-choice (MCQ) and open-ended evaluation formats (treated as binary: correct or incorrect). For statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for `wilcoxon` test; `pingouin`³ (Vallat, 2018) for intraclass correlation and `cronbach_alpha`; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), proving that the reported scores are precise estimates unaffected by random seeding noise.

Table 26: Model Accuracy and Variance across 3 Independent Runs.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The consistently low standard deviations across all models and evaluation formats establish that performance estimates are highly stable and reproducible, with minimal sensitivity to stochastic run-to-run variation. This stability proves that observed accuracy differences reflect underlying model capability rather than noise introduced by random initialization or decoding. Although open-ended evaluation exhibits slightly higher variance than MCQ, the absolute magnitude remains negligible, confirming that free-form generation is also robustly assessed under the current protocol. These findings demonstrate that the dataset size and scoring methodology yield convergent accuracy estimates without requiring extensive repeated runs. Single-run evaluations are therefore reliable for large-scale comparisons, while multi-run analyses serve to validate robustness. Overall, this establishes the empirical credibility of the reported results and supports confident cross-model and cross-format performance interpretation.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) Reliability Scores.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The consistently high ICC(3,k) scores demonstrate that the evaluation achieves near-perfect reliability across independent runs, regardless of model size or response format. This proves that item-level difficulty ordering is preserved and that performance fluctuations do not alter relative rankings among samples. Such strong agreement establishes that the scoring pipeline is internally consistent and insensitive to stochastic variation in model outputs. Critically, this rules out random guessing as a dominant factor, even in open-ended generation, where variability is typically higher. These

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

results confirm that the benchmark provides a dependable signal for assessing model capability. Longitudinal comparisons and fine-grained ablations can therefore be conducted with high confidence in measurement stability.

Overall, the combined evidence from multiple-run variance analysis and ICC reliability proves that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, confirming that results are not driven by random seeding or decoding noise. This robustness guarantees that observed performance gaps across models and formats reflect genuine capability differences. The evaluation framework therefore provides a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To guarantee statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides conclusive evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling Study (1,000 Trials): Wilcoxon Signed-Rank Test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-v1-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-v1-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results provide conclusive empirical evidence that accuracy estimates have already converged at relatively small subcategory sizes. The consistently high mean (p)-values and the absence of statistically significant differences across all 1,000 trials prove that performance metrics are insensitive to moderate reductions in sample size. This stability holds uniformly across models and evaluation formats, establishing that neither model scale nor output structure introduces additional sampling uncertainty. These findings demonstrate that the variance in accuracy is dominated by model behavior rather than data sparsity at the subcategory level. The current subcategory size (≈ 50) is therefore statistically redundant, offering a substantial safety margin beyond the minimum required for reliable estimation. This confirms the validity of the benchmark design and shows that future extensions can reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-Level Consistency Metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-v1-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-v1-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The extremely high mean pairwise correlations (> 0.94) demonstrate strong item-level stability, proving that the relative difficulty of individual questions is preserved across independent runs. This consistency establishes that model responses are systematically structured rather than subject to random variation, even under stochastic decoding. The similarly high Cronbach’s α values (> 0.9) confirm that the dataset exhibits excellent internal coherence, with items contributing reliably to the overall accuracy measure. The slight reductions observed in open-ended formats are expected given their greater expressive flexibility, yet remain well within the range of excellent reliability. Together, these results establish that the benchmark functions as a unified and internally consistent

measurement instrument. Model performance differences can therefore be confidently attributed to true capability variation rather than instability at the item level.

K.3.3 OVERALL DISCUSSION

Our expanded statistical testing proves that our evaluation and dataset are robust: **ICC and Cronbach's α scores consistently exceed 0.98**, and the **standard deviation of accuracy across runs is negligible**. The **resampling study** further confirms that even smaller subsets produce statistically indistinguishable results from the full set, proving that our subcategory sizes (~ 50) are sufficient for robust conclusions. These results establish that the granular analysis in the paper reflects genuine model capability differences, not statistical noise.

Our expanded statistical testing proves that the evaluation protocol and dataset are highly robust, with **ICC and Cronbach's α scores consistently exceeding 0.98**, establishing near-perfect reliability and internal consistency. The **standard deviation of accuracy across runs remains negligible** ($< 0.6\%$), confirming that reported model performance is stable and unaffected by stochastic effects such as random seeding or decoding variability. The **resampling study** demonstrates that even substantially smaller subsets ($N = 20$ and $N = 25$) yield statistically indistinguishable accuracy estimates compared to the full set, establishing that the chosen subcategory size (~ 25) sits well beyond the threshold required for convergence. Collectively, these results prove that both absolute scores and relative performance rankings are stable across runs and sampling regimes. This statistical robustness guarantees that fine-grained comparisons across models, formats, and subcategories are meaningful. The granular analyses reported in the paper therefore reflect genuine differences in model capability rather than artifacts of sampling variance or evaluation noise.